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(54) EJECTION ELEMENT SUBSTRATE AND LIQUID EJECTION HEAD

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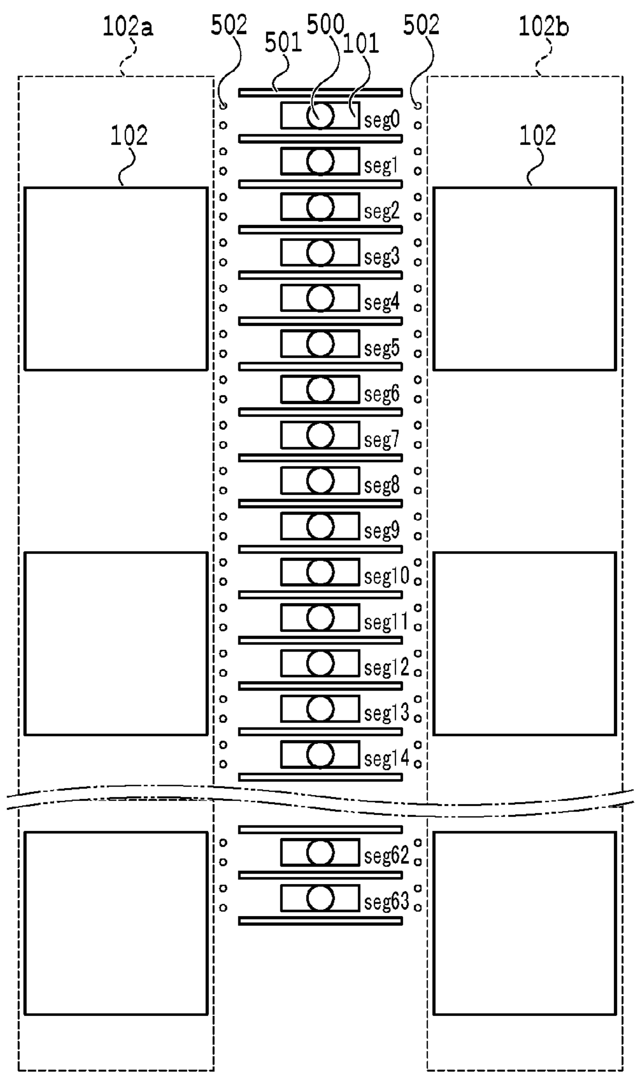
(57) ABSTRACT

An object is to reduce the area of an ejection element substrate. To this end, the ejection element substrate includes multiple n-th (n=1, 2) energy generation elements, multiple n-th drive circuits configured to drive the respective n-th energy generation elements, and an n-th selection circuit configured to select some of the n-th drive circuits from the multiple n-th drive circuits and cause the selected n-th drive circuits to drive some of the n-th energy generation elements corresponding to the selected n-th drive circuits, respectively. The multiple n-th drive circuits are arranged in a first direction, multiple first energy generation elements and multiple second energy generation elements are arranged side by side in a region between a region where multiple first drive circuits are arranged and a region where multiple second drive circuits are arranged such that a first direction is a main direction.



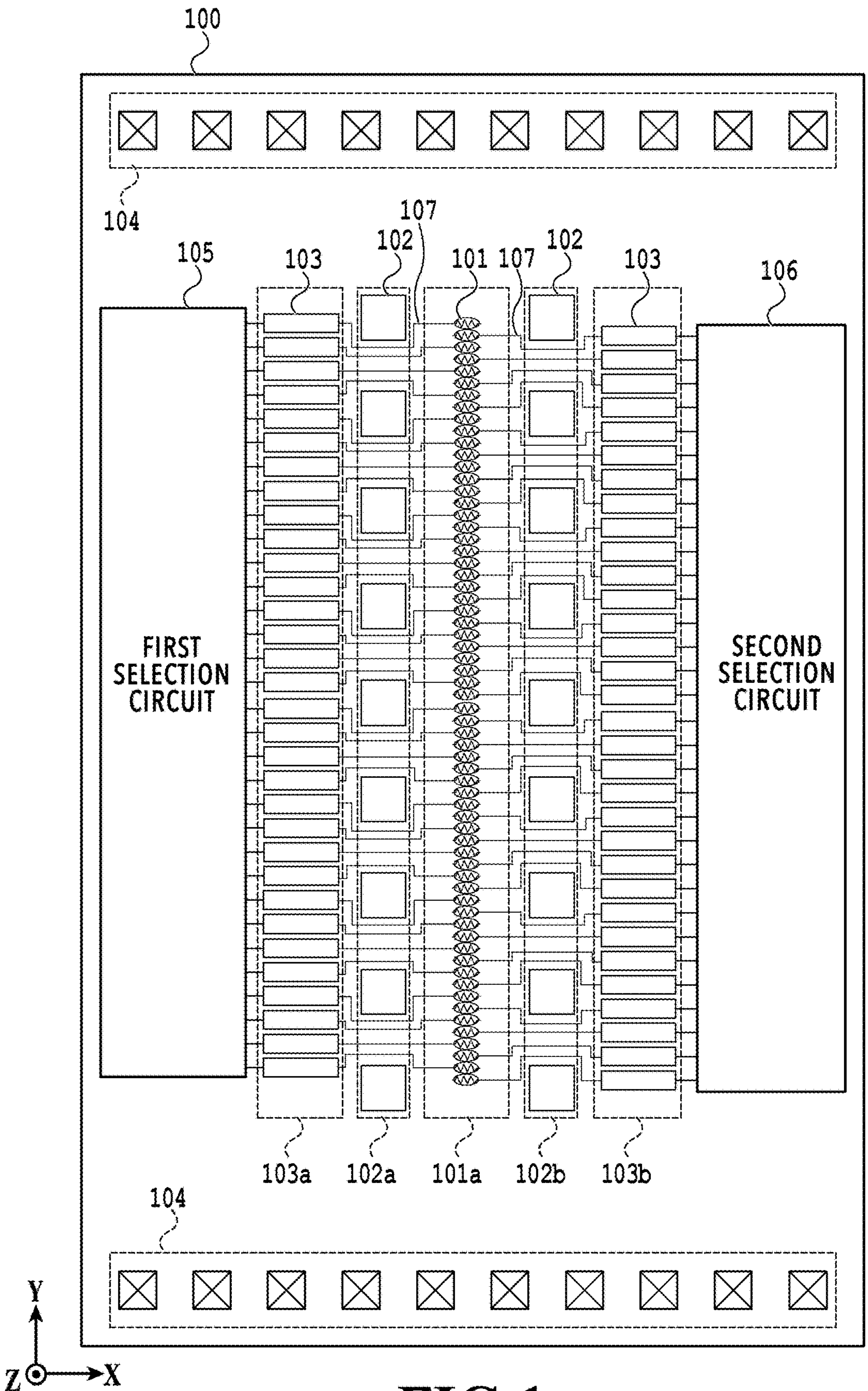

100
104
105
103
102
107
101
107
102
103
106
FIRST SELECTION CIRCUIT
SECOND SELECTION CIRCUIT
103a
102a
101a
102b
103b
104
Y
Z
X

FIG.1

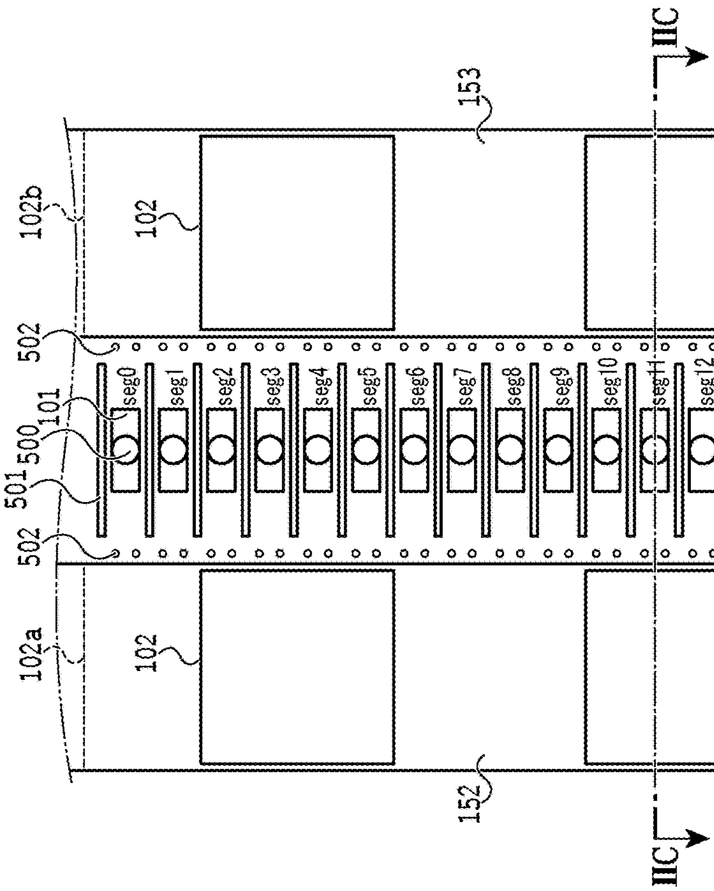


FIG.2B

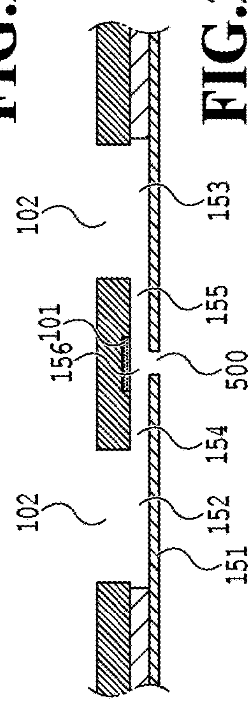


FIG.2C

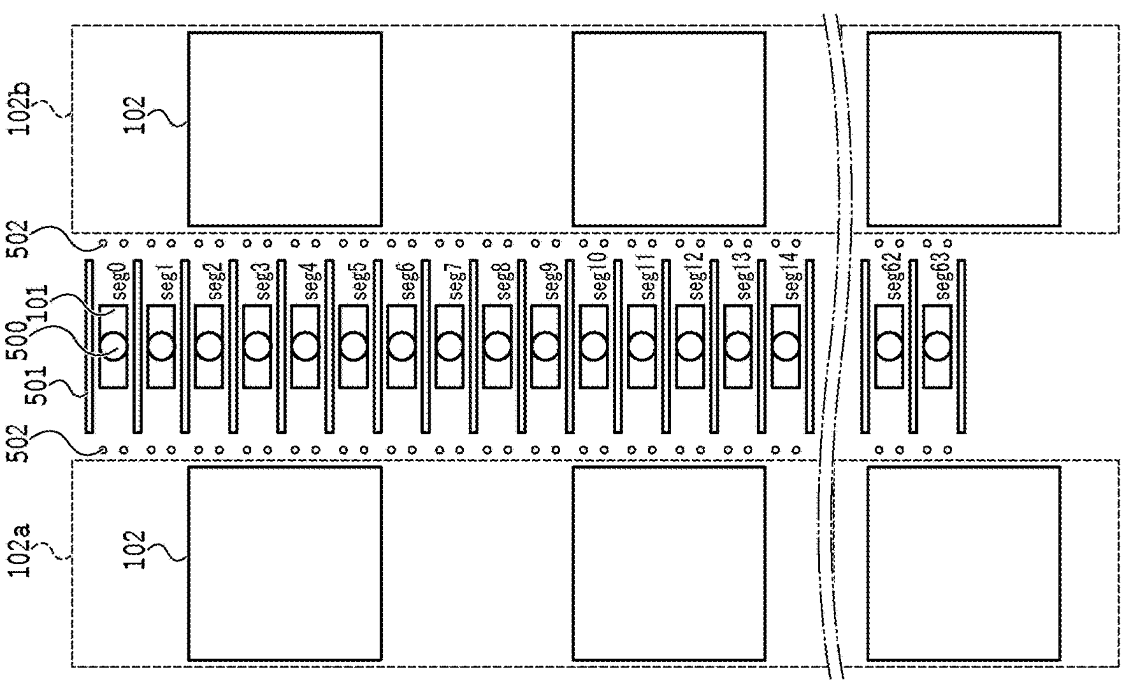
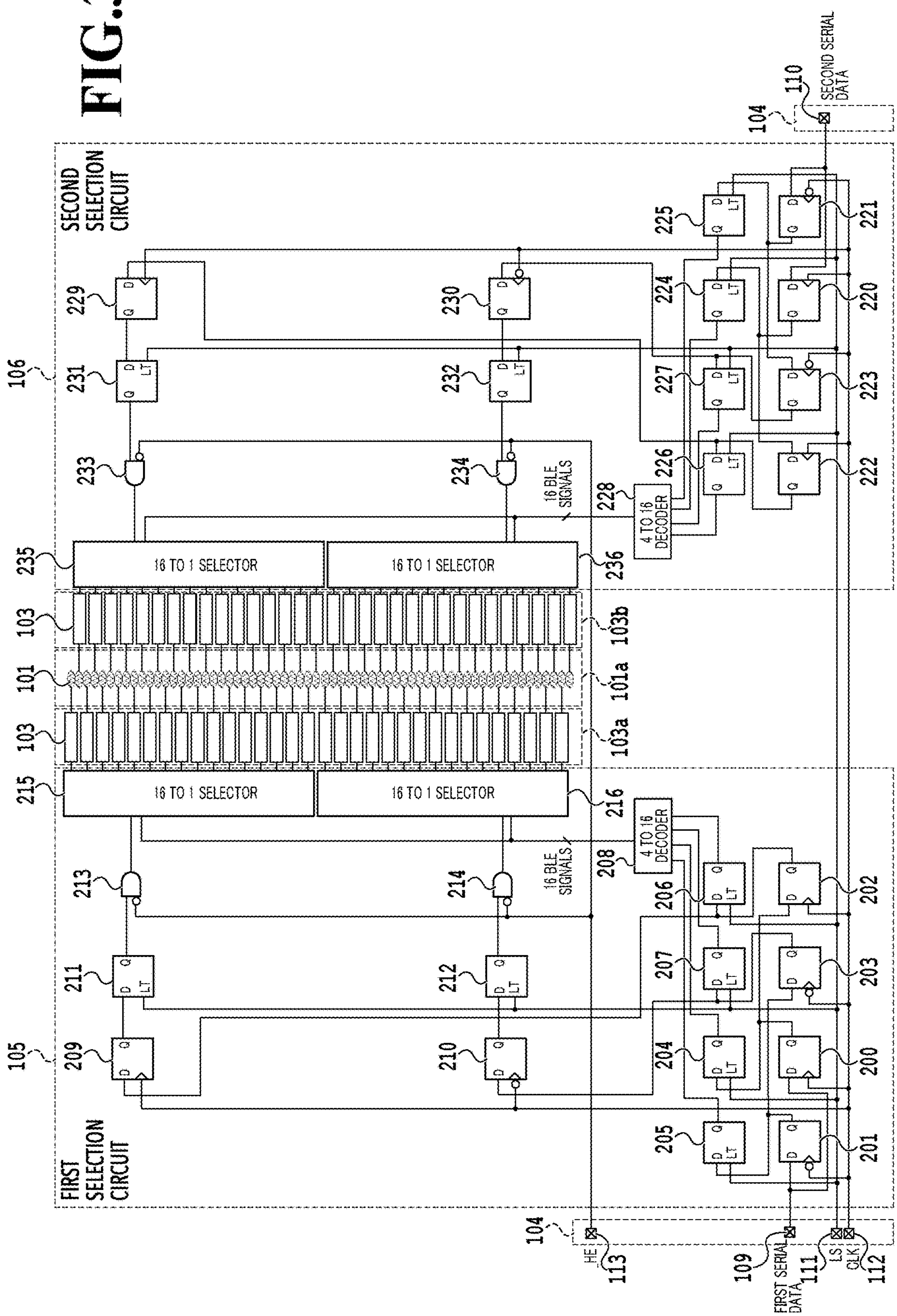


FIG.2A

FIG.3


SECOND SELECTION CIRCUIT
FIRST SELECTION CIRCUIT
16 TO 1 SELECTOR
16 TO 1 SELECTOR
16 TO 1 SELECTOR
16 TO 1 SELECTOR
4 TO 16 DECODER
4 TO 16 DECODER
16 BLE SIGNALS
16 BLE SIGNALS
SECOND SERIAL DATA
FIRST SERIAL DATA
HE
LS
CLK
101
101a
103
103a
103b
104
105
106
109
110
111
112
113
200
201
202
203
204
205
206
207
208
209
210
211
212
213
214
215
216
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221
222
223
224
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227
228
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232
233
234
235
236

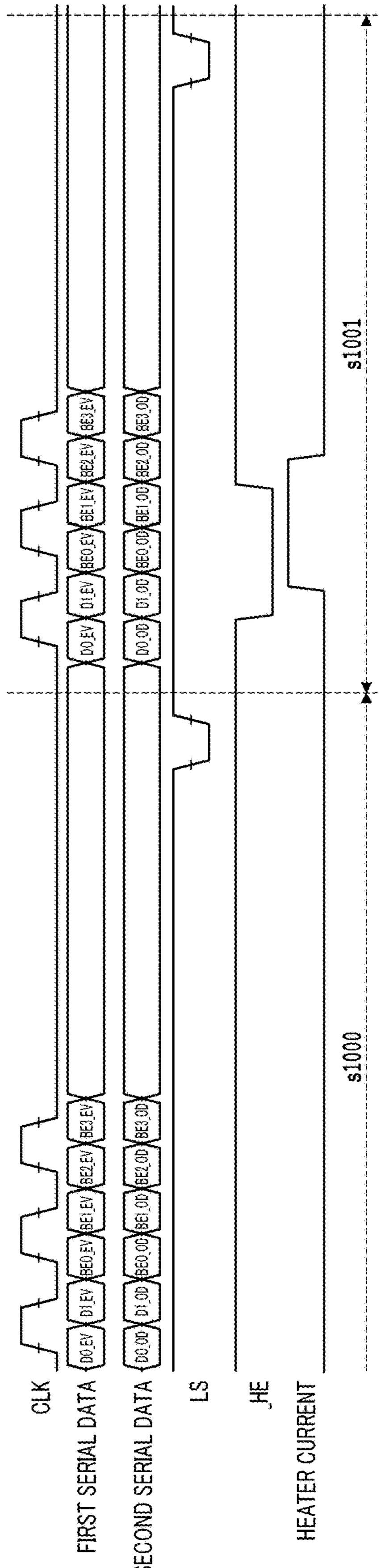

CLK
FIRST SERIAL DATA
SECOND SERIAL DATA
LS
HE
HEATER CURRENT
D0_EV
D1_EV
BE0_EV
BE1_EV
BE2_EV
BE3_EV
D0_OD
D1_OD
BE0_OD
BE1_OD
BE2_OD
BE3_OD
s1000
s1001

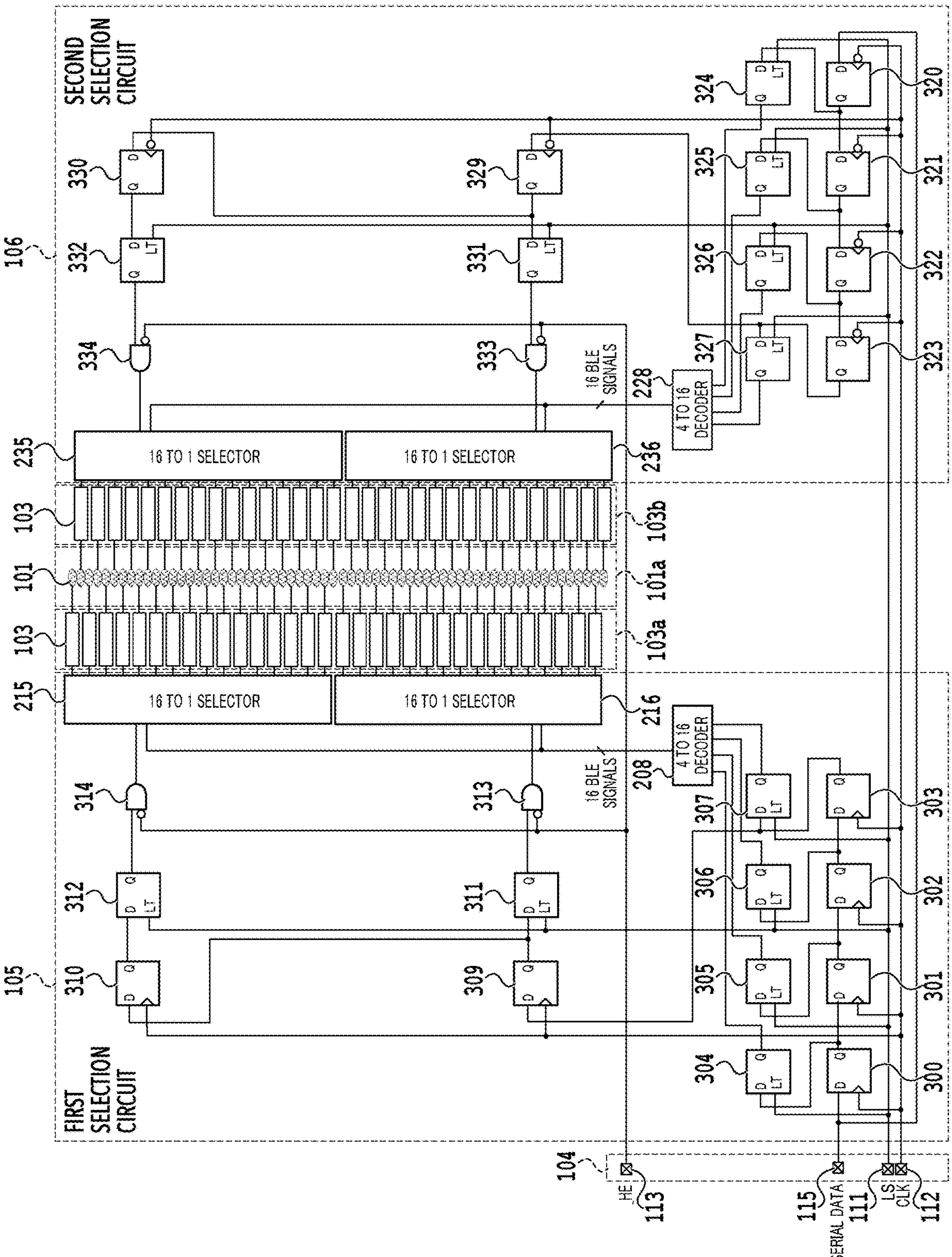
FIG.4

113 109 101 110

_HE	FIRST SERIAL DATA						Seg NUMBER	SECOND SERIAL DATA					
	BE3_EV	BE2_EV	BE1_EV	BE0_EV	D1_EV	D0_EV		D0_OD	D1_OD	BE0_OD	BE1_OD	BE2_OD	BE3_OD
Low	Low	Low	Low	Low	-	High	0						
Low							1	High	-	Low	Low	Low	Low
Low	Low	Low	Low	High	-	High	2						
Low							3	High	-	High	Low	Low	Low
Low	Low	Low	High	Low	-	High	4						
Low							5	High	-	Low	High	Low	Low
Low	Low	Low	High	High	-	High	6						
Low	High	Low	Low	High	-	High	18						
Low							19	High	-	High	Low	Low	High
Low	High	Low	High	Low	-	High	20						
Low							21	High	-	Low	High	Low	High
Low	High	Low	High	High	-	High	22						
Low													
Low	High	High	High	High	-	High	30						
Low							31	High	-	High	High	High	High
Low	Low	Low	Low	Low	High	-	32						
Low							33	-	High	Low	Low	Low	Low
Low	Low	Low	Low	High	High	-	34						
Low							35	-	High	High	Low	Low	Low
Low	Low	Low	High	Low	High	-	36						
Low							37	-	High	Low	High	Low	Low
Low	Low	Low	High	High	High	-	38						
Low							39	-	High	High	High	Low	Low
Low	High	Low	Low	Low	High	-	48						
Low							49	-	High	Low	Low	Low	High
Low	High	Low	Low	High	High	-	50						
Low							51	-	High	High	Low	Low	High
Low	High	Low	High	Low	High	-	52						
Low							53	-	High	Low	High	Low	High
Low	High	Low	High	High	High	-	54						
Low	High	High	High	High	High	-	62						
Low							63	-	High	High	High	High	High

FIG.5

FIG.6



FIRST SELECTION CIRCUIT
SECOND SELECTION CIRCUIT
105
106
104
16 TO 1 SELECTOR
16 TO 1 SELECTOR
16 TO 1 SELECTOR
16 TO 1 SELECTOR
215
216
235
236
103
103
101
103a
101a
103b
4 TO 16 DECODER
4 TO 16 DECODER
208
228
16 BLE SIGNALS
16 BLE SIGNALS
300
301
302
303
304
305
306
307
309
310
311
312
313
314
320
321
322
323
324
325
326
327
329
330
331
332
333
334
HE
SERIAL DATA
LS
CLK
113
115
111
112

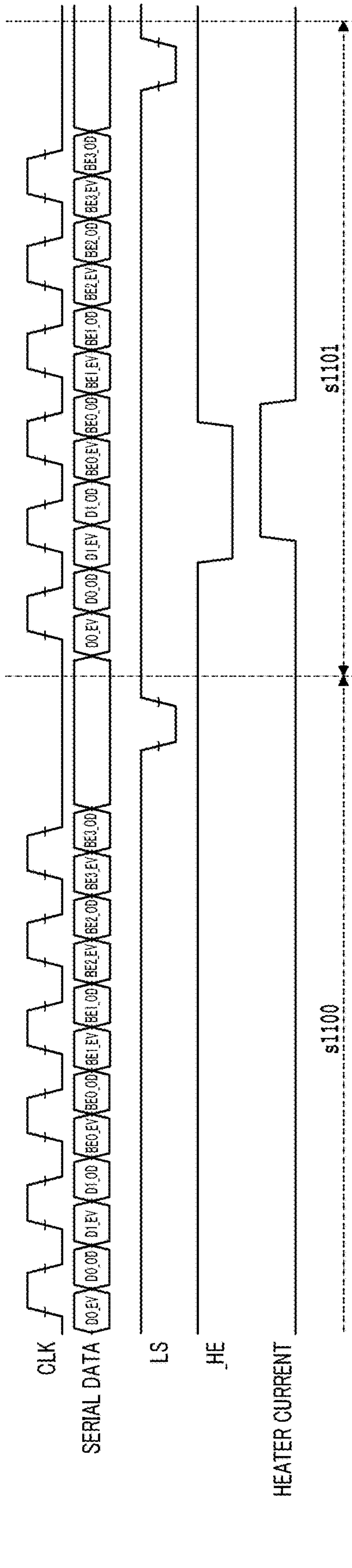

CLK
SERIAL DATA
LS
_HE
HEATER CURRENT
D0_EV
D0_OD
D1_EV
D1_OD
BE0_EV
BE0_OD
BE1_EV
BE1_OD
BE2_EV
BE2_OD
BE3_EV
BE3_OD
s1100
s1101

FIG. 7

113 115 101 115

_HE	SERIAL DATA						Seg NUMBER	SERIAL DATA					
	BE3_EV	BE2_EV	BE1_EV	BE0_EV	D1_EV	D0_EV		D0_OD	D1_OD	BE0_OD	BE1_OD	BE2_OD	BE3_OD
Low	Low	Low	Low	Low	-	High	0						
Low							1	High	-	Low	Low	Low	Low
Low	Low	Low	Low	High	-	High	2						
Low							3	High	-	High	Low	Low	Low
Low	Low	Low	High	Low	-	High	4						
Low							5	High	-	Low	High	Low	Low
Low	Low	Low	High	High	-	High	6						
Low	High	Low	Low	High	-	High	18						
Low							19	High	-	High	Low	Low	High
Low	High	Low	High	Low	-	High	20						
Low							21	High	-	Low	High	Low	High
Low	High	Low	High	High	-	High	22						
Low	High	High	High	High	-	High	30						
Low							31	High	-	High	High	High	High
Low	Low	Low	Low	Low	High	-	32						
Low							33	-	High	Low	Low	Low	Low
Low	Low	Low	Low	High	High	-	34						
Low							35	-	High	High	Low	Low	Low
Low	Low	Low	High	Low	High	-	36						
Low							37	-	High	Low	High	Low	Low
Low	Low	Low	High	High	High	-	38						
Low							39	-	High	High	High	Low	Low
Low	High	Low	Low	Low	High	-	48						
Low							49	-	High	Low	Low	Low	High
Low	High	Low	Low	High	High	-	50						
Low							51	-	High	High	Low	Low	High
Low	High	Low	High	Low	High	-	52						
Low							53	-	High	Low	High	Low	High
Low	High	Low	High	High	High	-	54						
Low	High	High	High	High	High	-	62						
Low							63	-	High	High	High	High	High

FIG.8

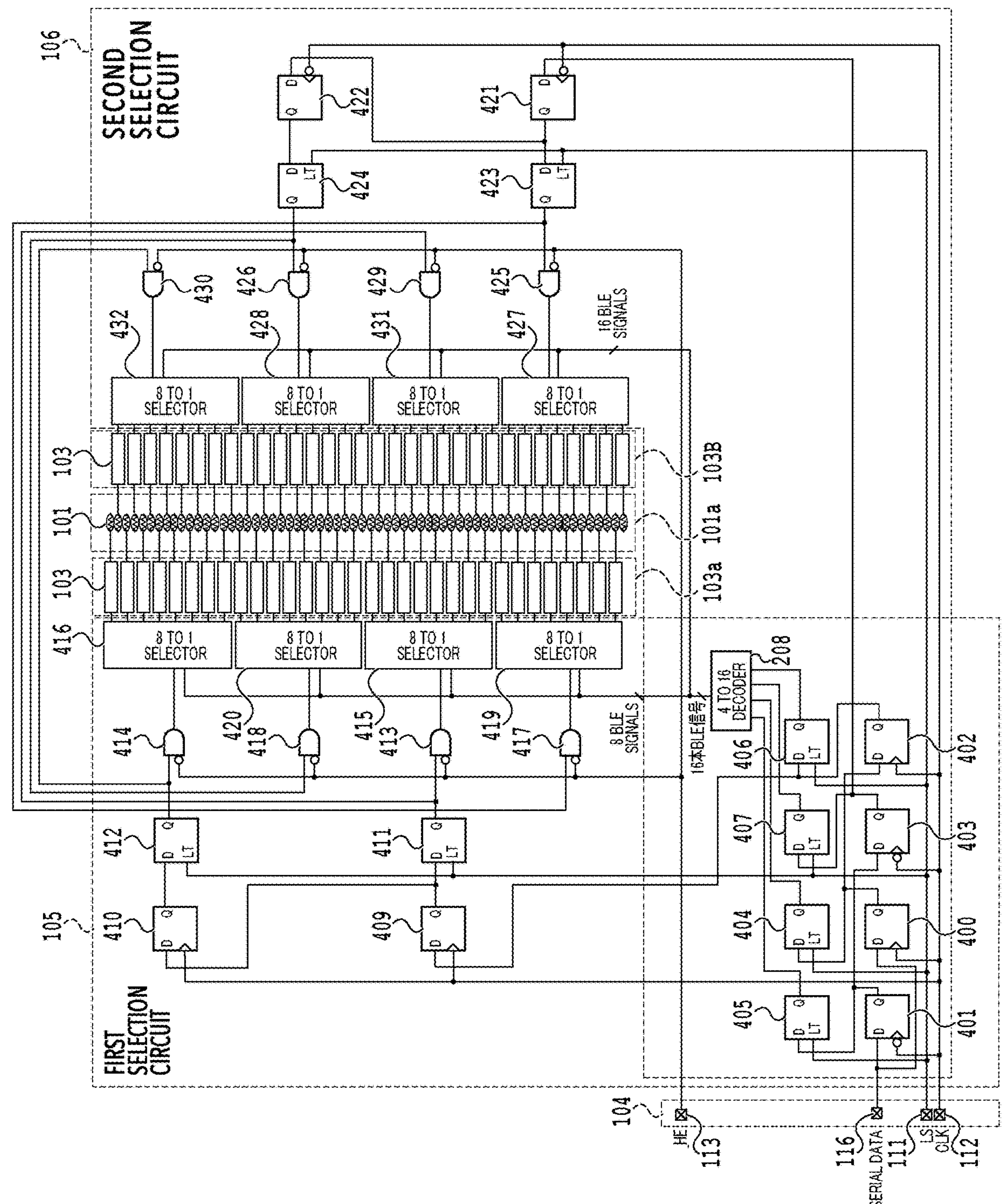

SECOND SELECTION CIRCUIT
FIRST SELECTION CIRCUIT
8 TO 1 SELECTOR
4 TO 16 DECODER
16 BLE SIGNALS
8 BLE SIGNALS
16本BLE信号
HE
SERIAL DATA
LS
CLK
101
101a
103
103a
103B
104
105
106
111
112
113
116
208
400
401
402
403
404
405
406
407
409
410
411
412
413
414
415
416
417
418
419
420
421
422
423
424
425
426
427
428
429
430
431
432

FIG.9

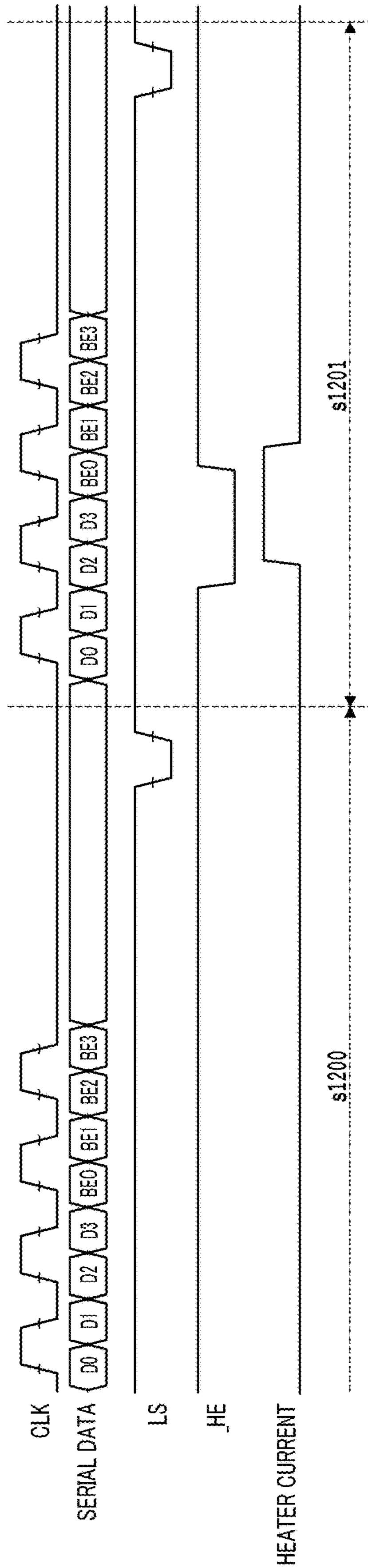

CLK
SERIAL DATA
LS
_HE
HEATER CURRENT
D0
D1
D2
D3
BE0
BE1
BE2
BE3
s1200
s1201

FIG.10

113 116 101

_HE	SERIAL DATA								Seg NUMBER
	BE3	BE2	BE1	BE0	D3	D2	D1	D0	
Low	Low	Low	Low	Low	–	–	–	High	0
Low	Low	Low	Low	High	–	–	–	High	1
Low	Low	Low	High	Low	–	–	–	High	2
Low	Low	Low	High	High	–	–	–	High	3
Low	Low	High	Low	Low	–	–	–	High	4
Low	Low	High	Low	High	–	–	–	High	5
Low	Low	High	High	Low	–	–	–	High	6
Low	Low	Low	High	Low	–	–	High	–	18
Low	Low	Low	High	High	–	–	High	–	19
Low	Low	High	Low	Low	–	–	High	–	20
Low	Low	High	Low	High	–	–	High	–	21
Low	Low	High	High	Low	–	–	High	–	22
Low	High	High	High	Low	–	–	High	–	30
Low	High	High	High	High	–	–	High	–	31
Low	Low	Low	Low	Low	–	High	–	–	32
Low	Low	Low	Low	High	–	High	–	–	33
Low	Low	Low	High	Low	–	High	–	–	34
Low	Low	Low	High	High	–	High	–	–	35
Low	Low	High	Low	Low	–	High	–	–	36
Low	Low	High	Low	High	–	High	–	–	37
Low	Low	High	High	Low	–	High	–	–	38
Low	Low	High	High	High	–	High	–	–	39
Low	Low	Low	Low	Low	High	–	–	–	48
Low	Low	Low	Low	High	High	–	–	–	49
Low	Low	Low	High	Low	High	–	–	–	50
Low	Low	Low	High	High	High	–	–	–	51
Low	Low	High	Low	Low	High	–	–	–	52
Low	Low	High	Low	High	High	–	–	–	53
Low	Low	High	High	Low	High	–	–	–	54
Low	High	High	High	Low	High	–	–	–	62
Low	High	High	High	High	High	–	–	–	63

FIG.11

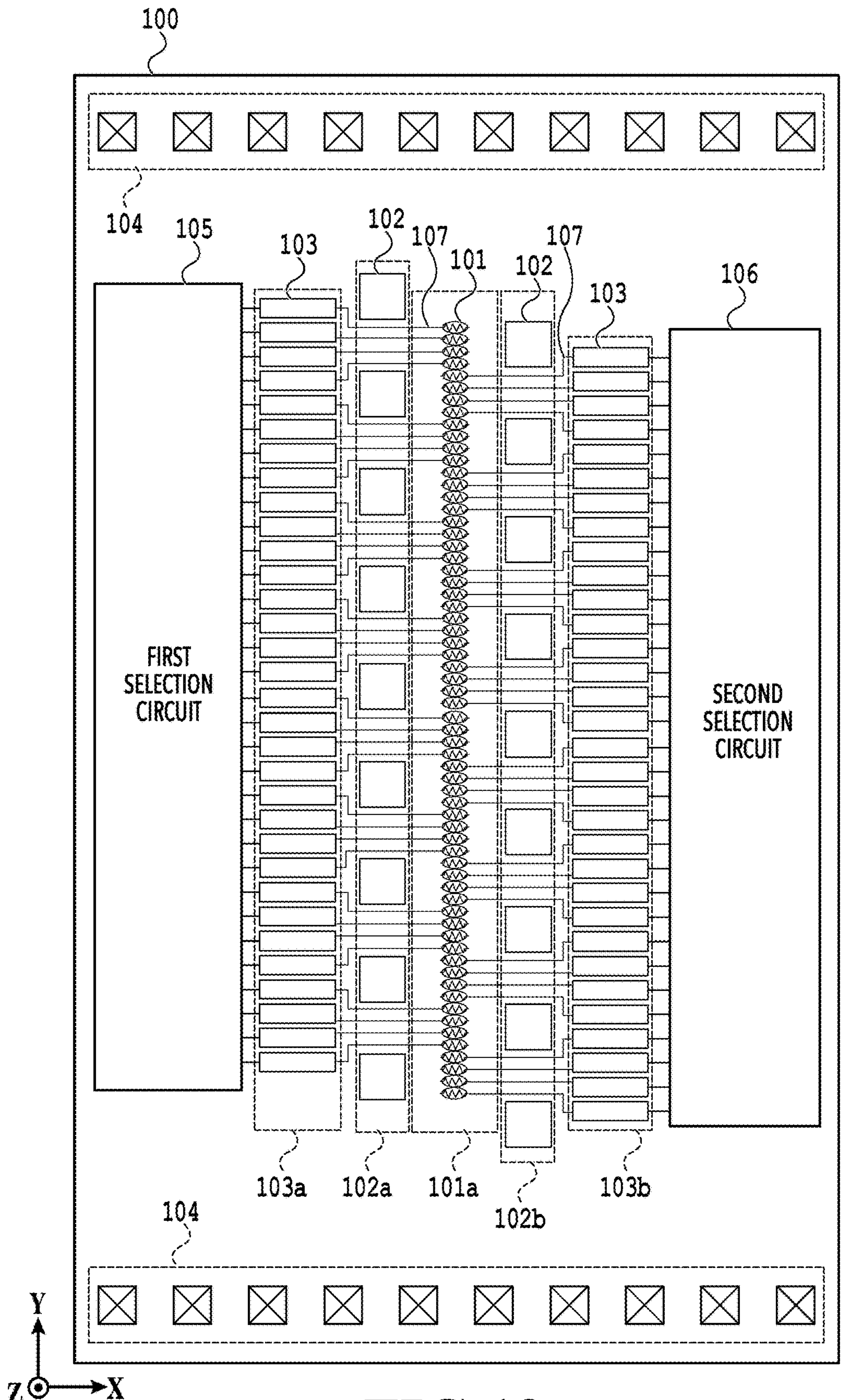

100
104
105
103
102
107
101
102
107
103
106
FIRST SELECTION CIRCUIT
SECOND SELECTION CIRCUIT
103a
102a
101a
102b
103b
104
Y
Z
X

FIG.12

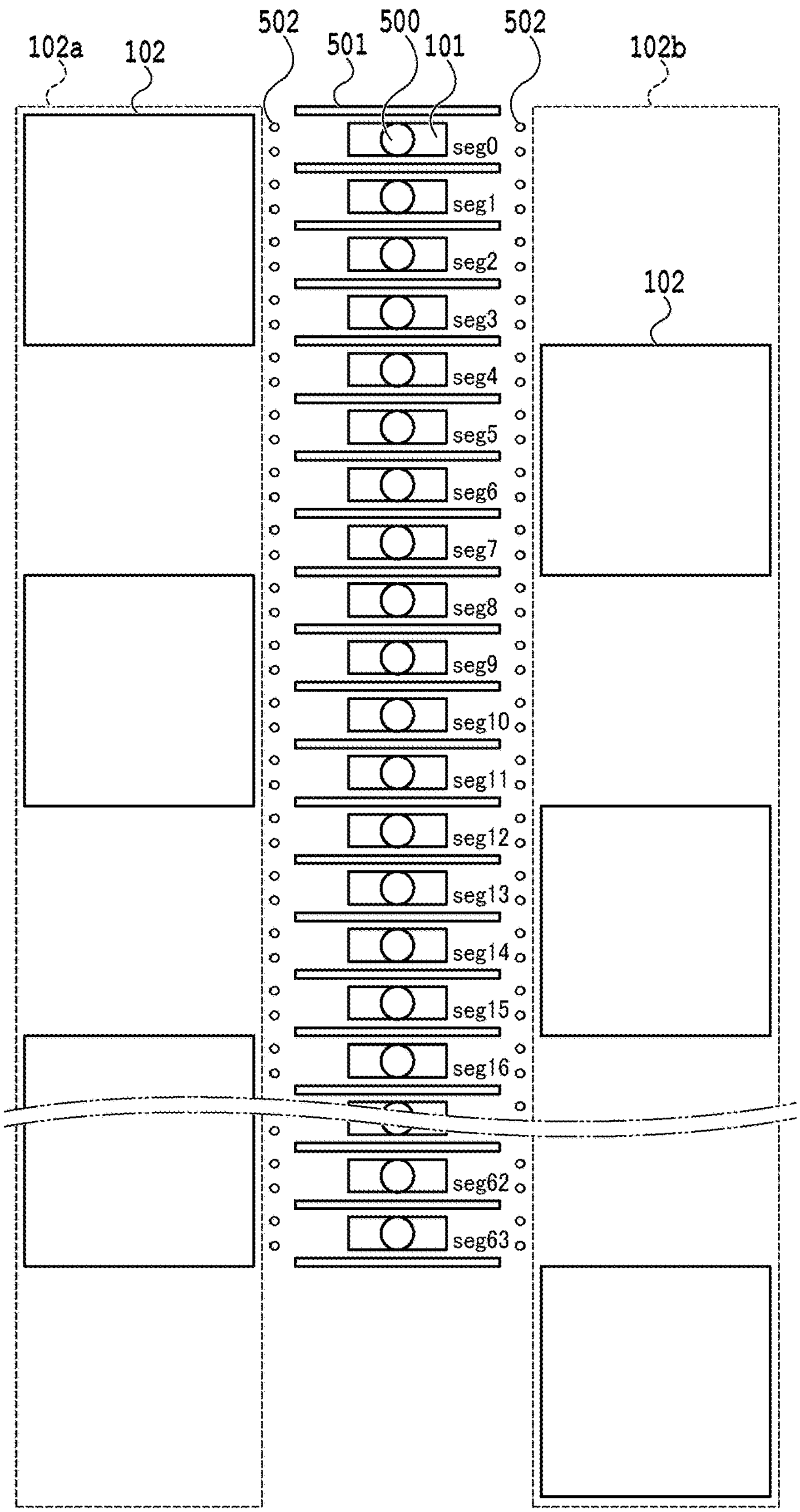
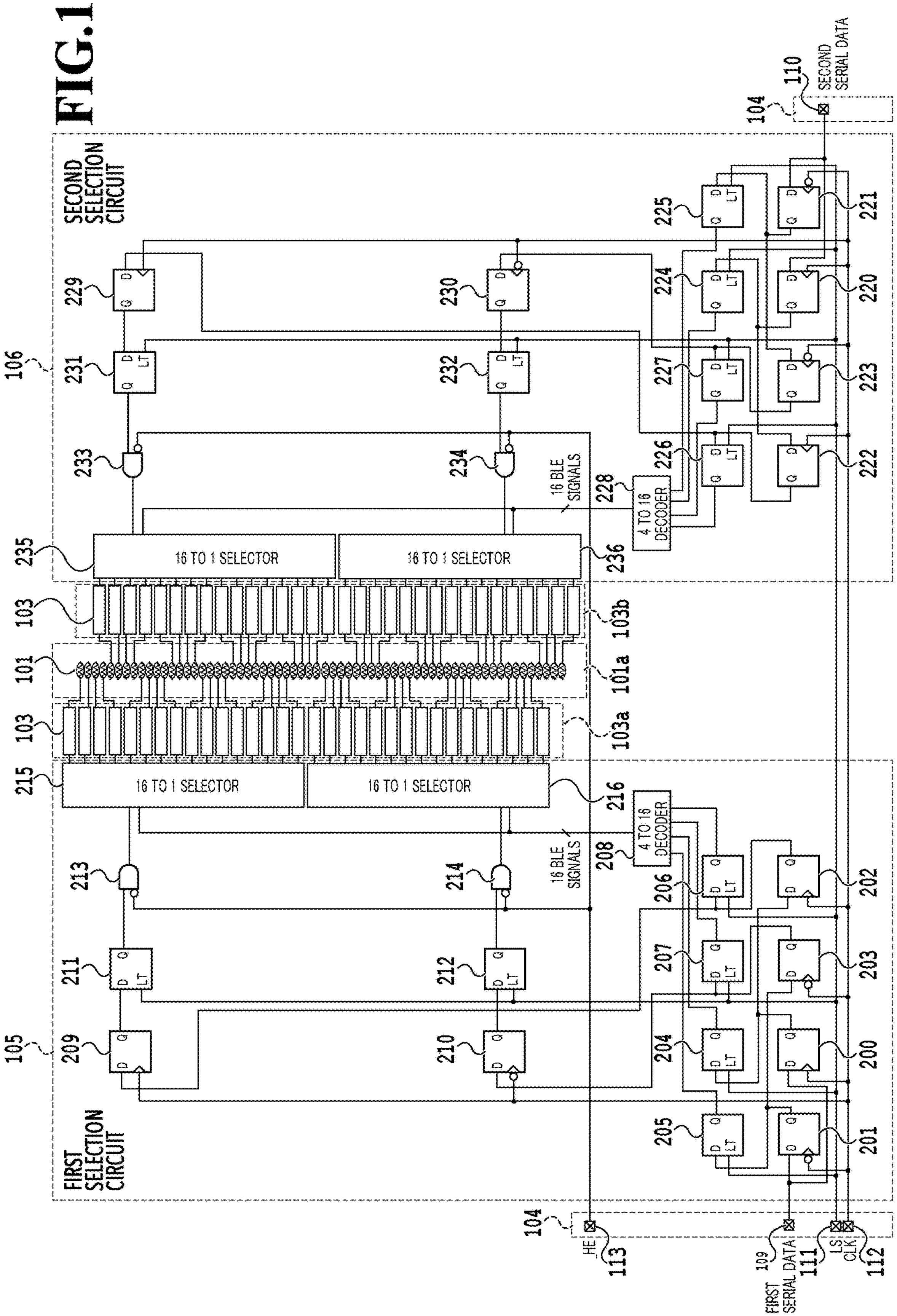

102a
102
502
501
500
101
502
102b
seg0
seg1
seg2
seg3
seg4
seg5
seg6
seg7
seg8
seg9
seg10
seg11
seg12
seg13
seg14
seg15
seg16
seg62
seg63
102

FIG.13

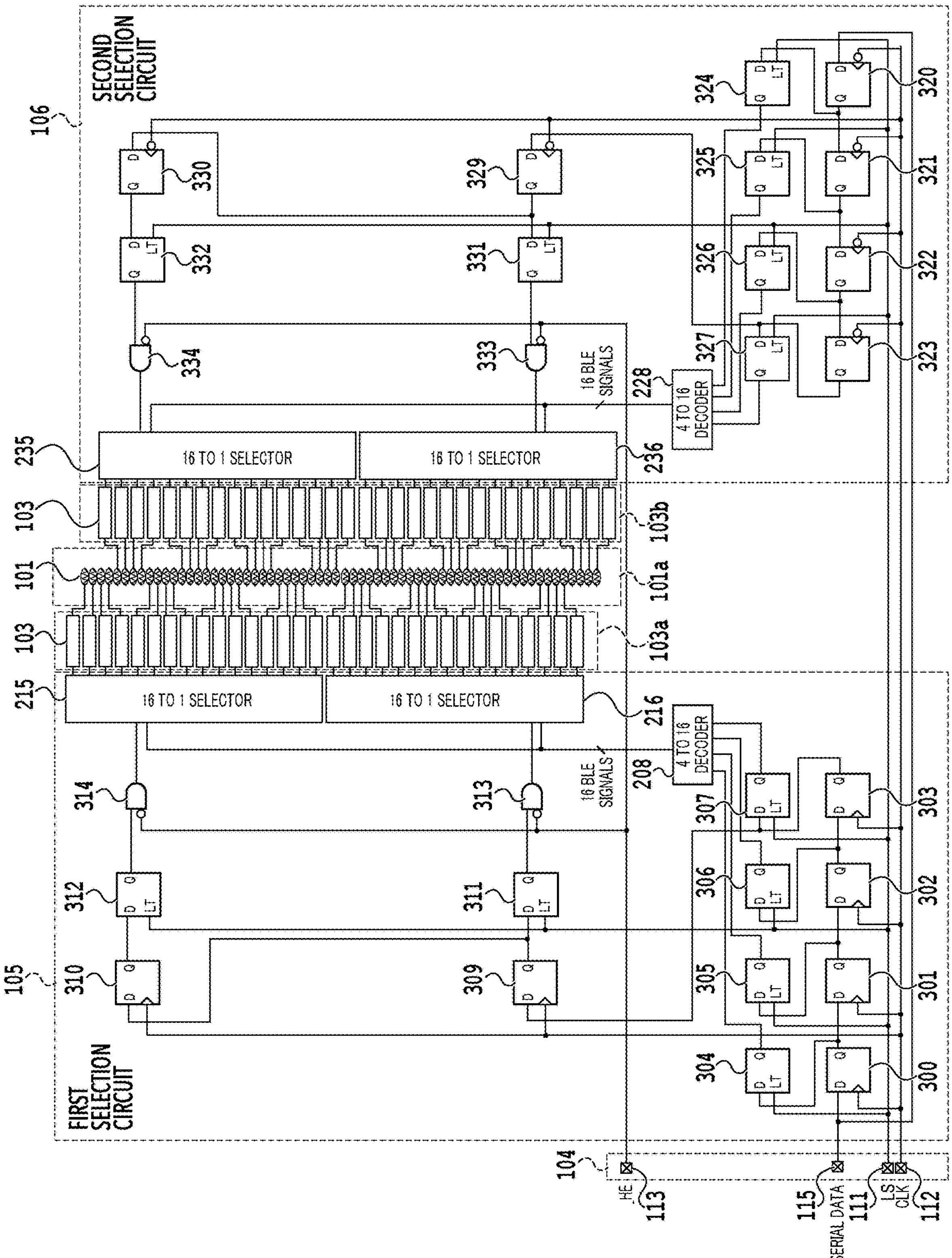
FIG.14


FIRST SELECTION CIRCUIT
SECOND SELECTION CIRCUIT
16 TO 1 SELECTOR
16 TO 1 SELECTOR
16 TO 1 SELECTOR
16 TO 1 SELECTOR
4 TO 16 DECODER
4 TO 16 DECODER
16 BLE SIGNALS
16 BLE SIGNALS
FIRST SERIAL DATA
SECOND SERIAL DATA
LS
CLK
_HE
D
Q
LT
101
101a
103
103a
103b
104
105
106
109
110
111
112
113
200
201
202
203
204
205
206
207
208
209
210
211
212
213
214
215
216
220
221
222
223
224
225
226
227
228
229
230
231
232
233
234
235
236

113 109 101 110

_HE	FIRST SERIAL DATA						Seg NUMBER	SECOND SERIAL DATA					
	BE3_EV	BE2_EV	BE1_EV	BE0_EV	D1_EV	D0_EV		D0_OD	D1_OD	BE0_OD	BE1_OD	BE2_OD	BE3_OD
Low	Low	Low	Low	Low	-	High	0						
Low	Low	Low	Low	High	-	High	1						
Low	Low	Low	High	Low	-	High	2						
Low	Low	Low	High	High	-	High	3						
Low							4	High	-	Low	Low	Low	Low
Low							5	High	-	High	Low	Low	Low
Low							6	High	-	Low	High	Low	Low
Low	High	Low	High	Low	-	High	18						
Low	High	Low	High	High	-	High	19						
Low							20	High	-	Low	Low	Low	High
Low							21	High	-	High	Low	Low	High
Low							22	High	-	Low	High	Low	High
Low							30	High	-	Low	High	High	High
Low							31	High	-	High	High	High	High
Low	Low	Low	Low	Low	High	-	32						
Low	Low	Low	Low	High	High	-	33						
Low	Low	Low	High	Low	High	-	34						
Low	Low	Low	High	High	High	-	35						
Low							36	-	High	Low	Low	Low	Low
Low							37	-	High	High	Low	Low	Low
Low							38	-	High	Low	High	Low	Low
Low							39	-	High	High	High	Low	Low
Low	High	Low	Low	High	High	-	49						
Low	High	Low	High	Low	High	-	50						
Low	High	Low	High	High	High	-	51						
Low							52	-	High	Low	Low	Low	High
Low							53	-	High	High	Low	Low	High
Low							54	-	High	Low	High	Low	High
Low							61	-	High	High	Low	High	High
Low							62	-	High	Low	High	High	High
Low							63	-	High	High	High	High	High

FIG.15


SECOND SELECTION CIRCUIT
FIRST SELECTION CIRCUIT
16 TO 1 SELECTOR
16 TO 1 SELECTOR
16 TO 1 SELECTOR
16 TO 1 SELECTOR
4 TO 16 DECODER
4 TO 16 DECODER
16 BLE SIGNALS
16 BLE SIGNALS
HE
SERIAL DATA
LS
CLK
106
105
235
236
215
216
228
208
103
103
103a
103b
101
101a
104
113
115
111
112
300
301
302
303
304
305
306
307
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314
320
321
322
323
324
325
326
327
329
330
331
332
333
334

FIG.16

113 115 101 115

_HE	SERIAL DATA						Seg NUMBER	SERIAL DATA					
	BE3_EV	BE2_EV	BE1_EV	BE0_EV	D1_EV	D0_EV		D0_OD	D1_OD	BE0_OD	BE1_OD	BE2_OD	BE3_OD
Low	Low	Low	Low	Low	-	High	0						
Low	Low	Low	Low	High	-	High	1						
Low	Low	Low	High	Low	-	High	2						
Low	Low	Low	High	High	-	High	3						
Low							4	High	-	Low	Low	Low	Low
Low							5	High	-	High	Low	Low	Low
Low							6	High	-	Low	High	Low	Low
Low	High	Low	High	Low	-	High	18						
Low	High	Low	High	High	-	High	19						
Low							20	High	-	Low	Low	Low	High
Low							21	High	-	High	Low	Low	High
Low							22	High	-	Low	High	Low	High
Low							30	High	-	Low	High	High	High
Low							31	High	-	High	High	High	High
Low	Low	Low	Low	Low	High	-	32						
Low	Low	Low	Low	High	High	-	33						
Low	Low	Low	High	Low	High	-	34						
Low	Low	Low	High	High	High	-	35						
Low							36	-	High	Low	Low	Low	Low
Low							37	-	High	High	Low	Low	Low
Low							38	-	High	Low	High	Low	Low
Low							39	-	High	High	High	Low	Low
Low	High	Low	Low	Low	High	-	48						
Low	High	Low	Low	High	High	-	49						
Low	High	Low	High	Low	High	-	50						
Low	High	Low	High	High	High	-	51						
Low							52	-	High	Low	Low	Low	High
Low							53	-	High	High	Low	Low	High
Low							54	-	High	Low	High	Low	High
Low							62	-	High	Low	High	High	High
Low							63	-	High	High	High	High	High

FIG.17

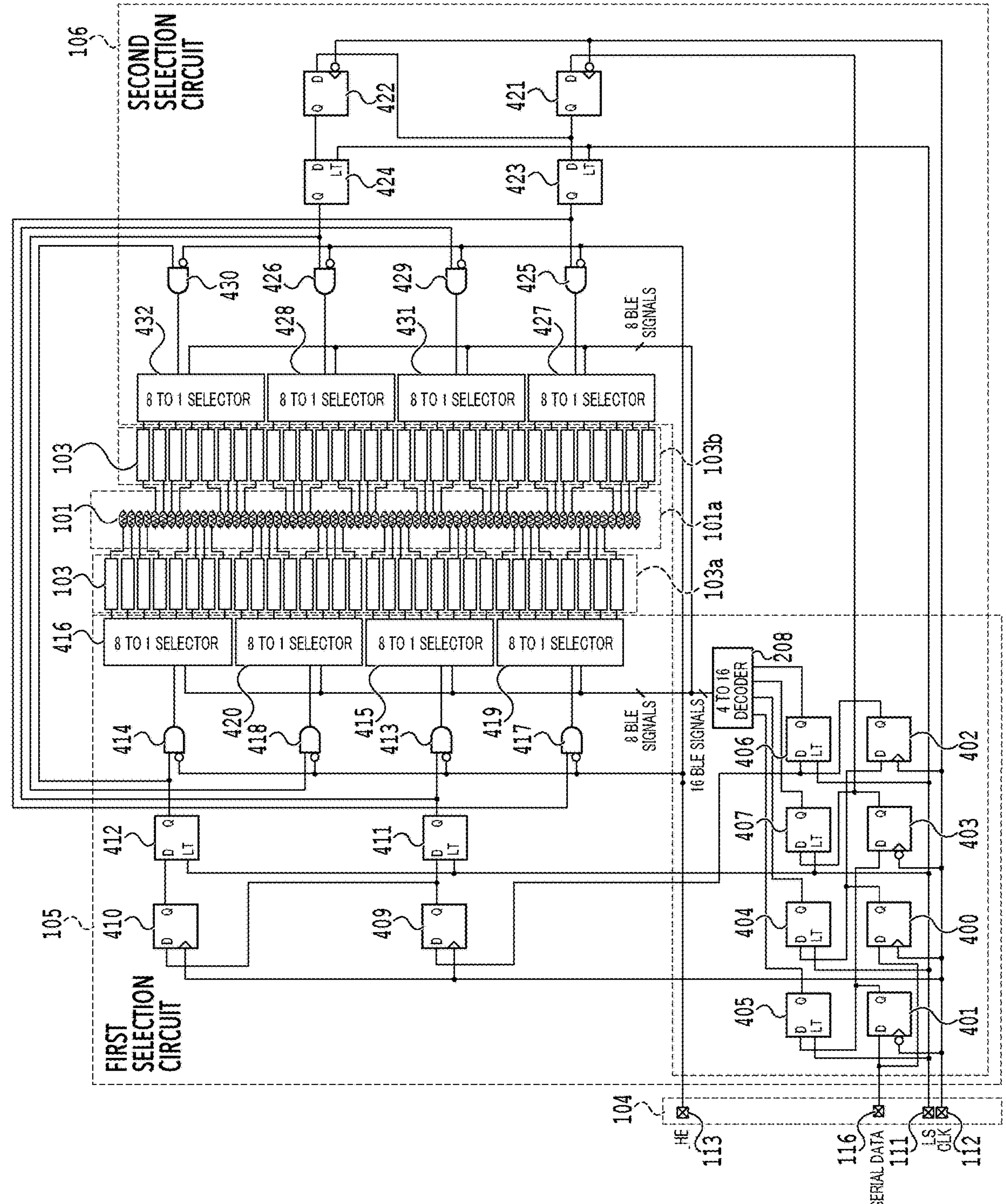
FIRST SELECTION CIRCUIT
SECOND SELECTION CIRCUIT
8 TO 1 SELECTOR
4 TO 16 DECODER
8 BLE SIGNALS
16 BLE SIGNALS
HE
SERIAL DATA
LS
CLK
101
101a
103
103a
103b
104
105
106
111
112
113
116
208
400
401
402
403
404
405
406
407
409
410
411
412
413
414
415
416
417
418
419
420
421
422
423
424
425
426
427
428
429
430
431
432

FIG.18

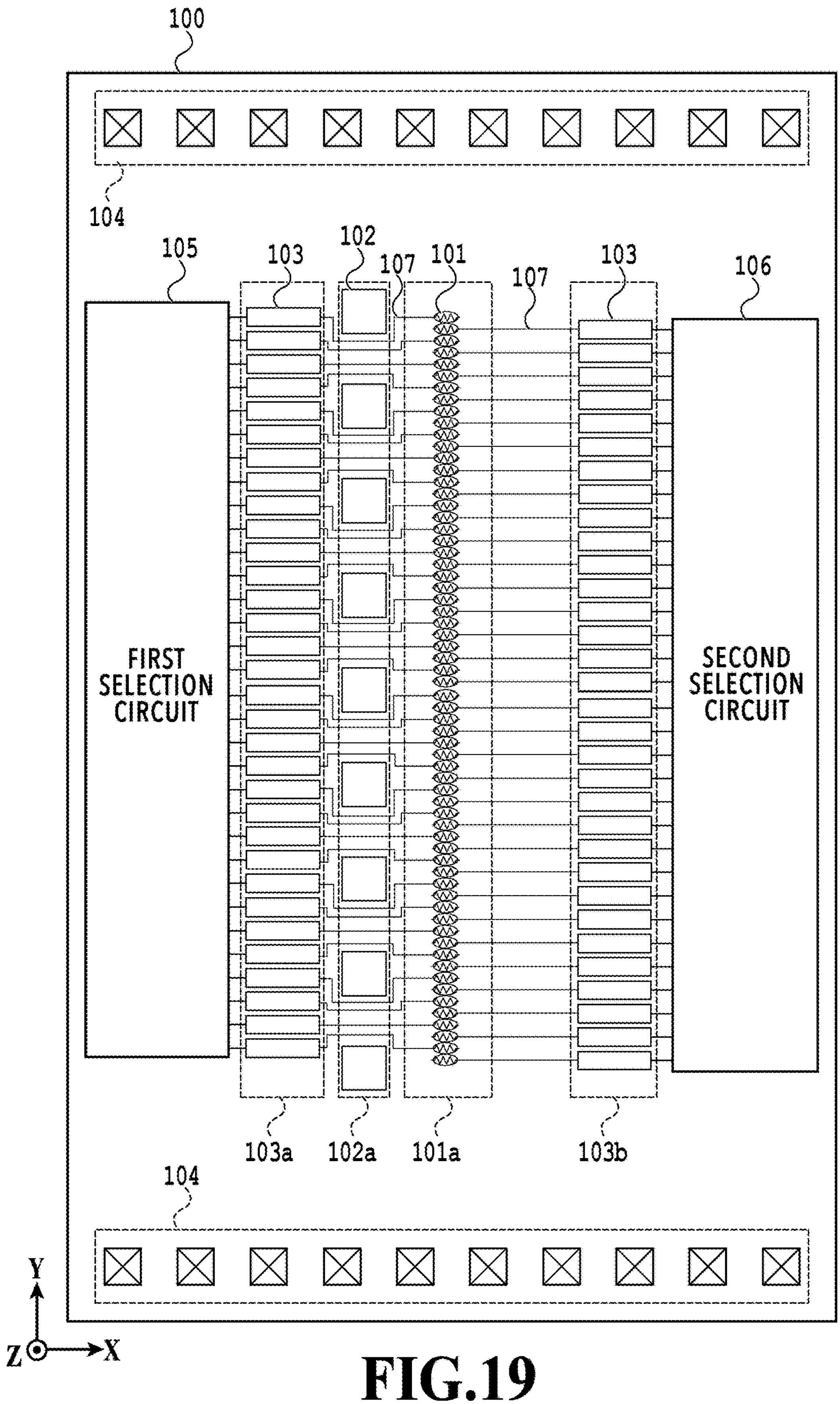
100
104
105
103
102
107
101
107
103
106
FIRST SELECTION CIRCUIT
SECOND SELECTION CIRCUIT
103a
102a
101a
103b
104
Y
Z
X

FIG.19

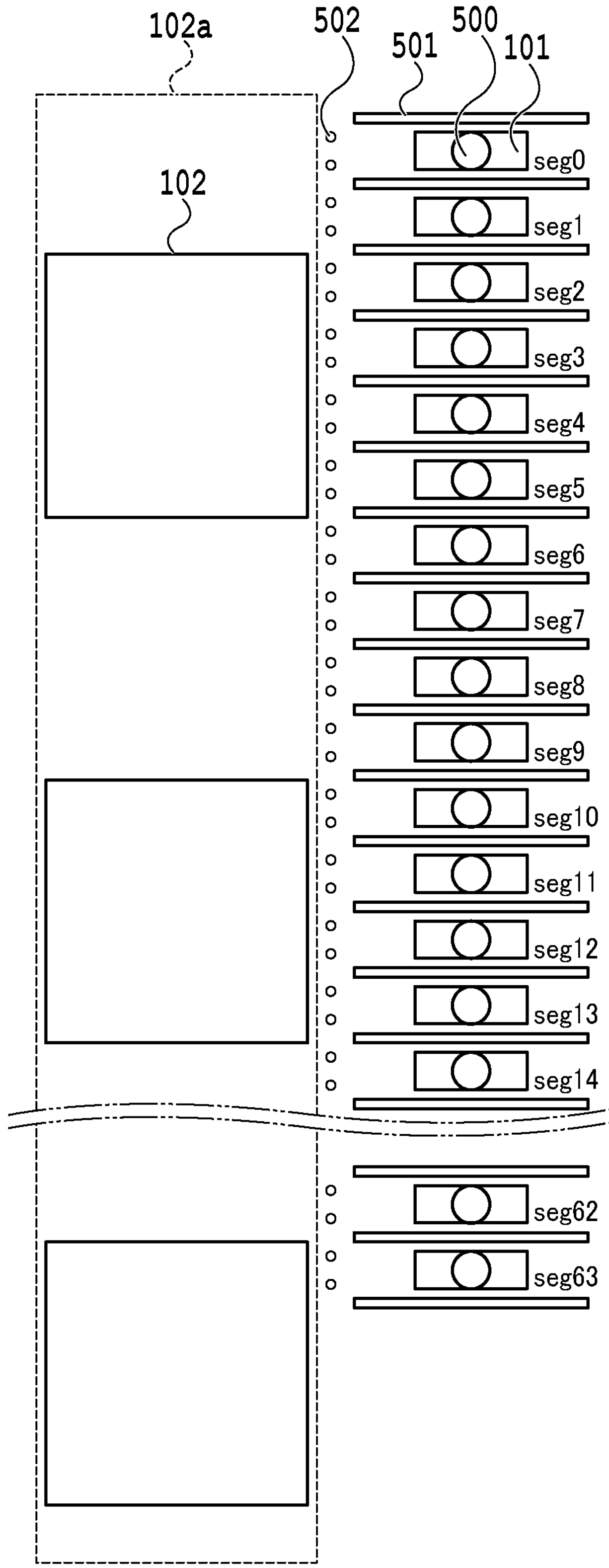
102a
502
500
501
101
102
seg0
seg1
seg2
seg3
seg4
seg5
seg6
seg7
seg8
seg9
seg10
seg11
seg12
seg13
seg14
seg62
seg63

FIG.20

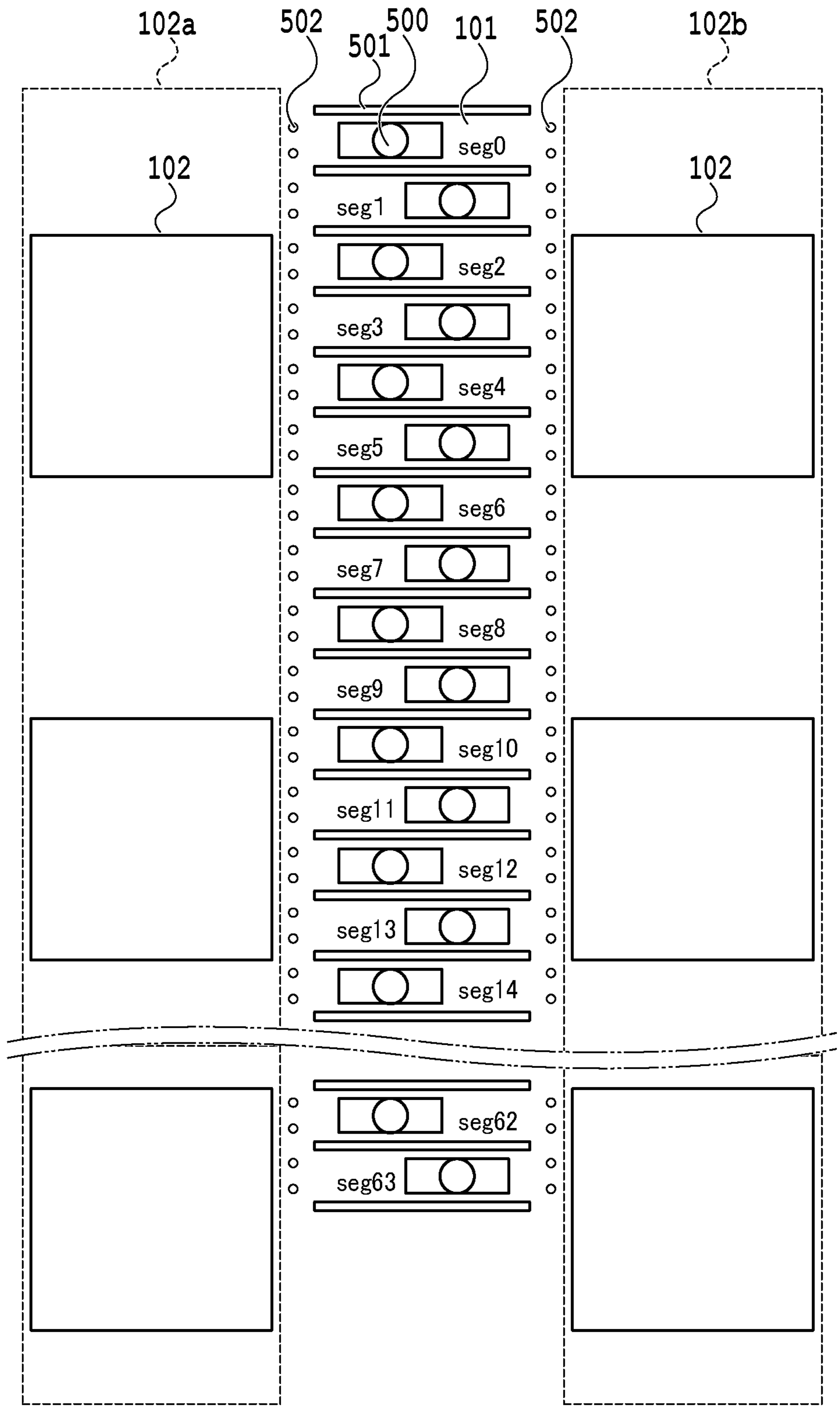
102a
502
501
500
101
502
102b
102
102
seg0
seg1
seg2
seg3
seg4
seg5
seg6
seg7
seg8
seg9
seg10
seg11
seg12
seg13
seg14
seg62
seg63

FIG.21

EJECTION ELEMENT SUBSTRATE AND LIQUID EJECTION HEAD

BACKGROUND

Field

[0001] The present disclosure relates to an ejection element substrate and a liquid ejection head.

Description of the Related Art

[0002] A liquid ejection head used in a liquid ejection apparatus such as an inkjet printer uses an ejection element substrate as one of components. Multiple liquid ejection ports for ejecting liquid are arranged side by side on a surface of an orifice plate attached to the ejection element substrate to penetrate the orifice plate. Each liquid ejection port communicates with a pressure chamber provided in a boundary region between the ejection element substrate and the orifice plate. The liquid in the pressure chamber is ejected from the liquid ejection port to a printing medium such as a sheet by being driven by an energy generation element. Japanese Patent Laid-Open No. 2012-254527 discloses an ejection element substrate in which liquid ejection ports are arranged in a staggered pattern of two arrays to increase the number of ink ejection ports in an array direction.

[0003] However, in the ejection element substrate disclosed in Japanese Patent Laid-Open No. 2012-254527, as illustrated in FIG. **1** thereof, multiple ink supply ports arranged side by side in two arrays and multiple through-holes arranged side by side in one array are arranged between the two arrays of the liquid ejection ports. Accordingly, a distance between the two arrays of the liquid ejection port is large. This increases the area of the ejection element substrate, and may become a hinderance in downsizing of the liquid ejection head.

SUMMARY

[0004] The present disclosure has been made in view of the above-mentioned points, and an object is to reduce the area of an ejection element substrate.

[0005] In a first aspect of the present disclosure, there is provided An ejection element substrate comprising: a plurality of first energy generation elements; a plurality of first drive circuits configured to drive the plurality of first energy generation elements, respectively; a first selection circuit configured to select some of the first drive circuits from the plurality of first drive circuits and cause the selected first drive circuits to drive some of the first energy generation elements corresponding to the selected first drive circuits, respectively; a plurality of second energy generation elements; a plurality of second drive circuits configured to drive the plurality of second energy generation elements, respectively; and a second selection circuit configured to select some of the second drive circuits from the plurality of second drive circuits and cause the selected second drive circuits to drive some of the second energy generation elements corresponding to the selected second drive circuits, respectively, wherein the plurality of first drive circuits are arranged side by side in a first direction and the plurality of second drive circuits are arranged side by side along the first direction, and the plurality of first energy generation elements and the plurality of second energy generation elements are arranged side by side in a region between a region where the plurality of first drive circuits are arranged and a region where the plurality of second drive circuits are arranged such that a set of an integer number of the first energy generation elements and a set of the integer number of the second energy generation elements are arranged alternately with the first direction being a main direction, the integer number being one or more.

[0006] In a second aspect of the present disclosure, there is provided A liquid ejection head comprising: an ejection element substrate; and an orifice plate attached to the ejection element substrate, wherein the ejection element substrate includes: a plurality of first energy generation elements; a plurality of first drive circuits configured to drive the plurality of first energy generation elements, respectively; a first selection circuit configured to select some of the first drive circuits from the plurality of first drive circuits and cause the selected first drive circuits to drive some of the first energy generation elements corresponding to the selected the first drive circuits, respectively; a plurality of second energy generation elements; a plurality of second drive circuits configured to drive the plurality of second energy generation elements, respectively; and a second selection circuit configured to select some of the second drive circuits from the plurality of second drive circuits and cause the selected second drive circuits to drive some of the second energy generation elements corresponding to the selected second drive circuits, respectively, the plurality of first drive circuits are arranged side by side along a first direction, and the plurality of second drive circuits are arranged side by side along the first direction, the plurality of first energy generation elements and the plurality of second energy generation elements are arranged side by side in a region between a region where the plurality of first drive circuits are arranged and a region where the plurality of second drive circuits are arranged such that the first direction is a main direction, a boundary portion between the ejection element substrate and the orifice plate is provided with a first common flow path communicating with a plurality of first ink supply ports, a second common flow path communicating with a plurality of second ink supply ports, a plurality of first individual flow paths communicating with the first common flow path, a plurality of second individual flow paths communicating with the second common flow path, and a plurality of pressure chambers communicating with the plurality of first individual flow paths, respectively, and with the plurality of second individual flow paths, respectively, the orifice plate includes a plurality of ejection ports that allow the plurality of pressure chambers to communicate with an outside, and the energy generation elements are arranged at positions where the energy generation elements face the ejection ports of the pressure chambers, respectively.

[0007] Further features of the present disclosure will become apparent from the following description of exemplary embodiments with reference to the attached drawings.

BRIEF DESCRIPTION OF THE DRAWINGS

[0008] FIG. **1** is a plan diagram illustrating an entire ejection element substrate according to first to third embodiments;

[0009] FIG. **2**A is a partially-enlarged plan diagram illustrating part of the ejection element substrate according to the first to third embodiments;

[0010] FIG. **2**B is a partially-enlarged plan diagram illustrating part of a liquid ejection head according to the first to third embodiments;

[0011] FIG. **2**C is a cross-sectional diagram of the liquid ejection head according to the first to third embodiments;

[0012] FIG. **3** is a circuit diagram illustrating circuits mounted in the ejection element substrate according to the first embodiment;

[0013] FIG. **4** is a timing chart illustrating an operation of the circuits mounted in the ejection element substrate according to the first embodiment;

[0014] FIG. **5** illustrates a table illustrating a relationship between selection control data to be inputted into two selection circuits and heaters to be selected by these selection circuits in the ejection element substrate according to the first embodiment;

[0015] FIG. **6** is a circuit diagram illustrating circuits mounted in the ejection element substrate according to the second embodiment;

[0016] FIG. **7** is a timing chart illustrating an operation of the circuits mounted in the ejection element substrate according to the second embodiment;

[0017] FIG. **8** illustrates a table illustrating a relationship between selection control data to be inputted into two selection circuits and heaters to be selected by these selection circuits in the ejection element substrate according to the second embodiment;

[0018] FIG. **9** is a circuit diagram illustrating circuits mounted in the ejection element substrate according to the third embodiment;

[0019] FIG. **10** illustrates a timing chart illustrating an operation of the circuits mounted in the ejection element substrate according to the third embodiment;

[0020] FIG. **11** illustrates a table illustrating a relationship between selection control data to be inputted into two selection circuits and heaters to be selected by these selection circuits in the ejection element substrate according to the third embodiment;

[0021] FIG. **12** is a plan diagram illustrating an entire ejection element substrate according to fourth to sixth embodiments;

[0022] FIG. **13** is a partially-enlarged plan diagram illustrating part of the ejection element substrate according to the fourth to sixth embodiments;

[0023] FIG. **14** is a circuit diagram illustrating circuits mounted in the ejection element substrate according to the fourth embodiment;

[0024] FIG. **15** illustrates a table illustrating a relationship between selection control data to be inputted into two selection circuits and heaters to be selected by these selection circuits in the ejection element substrate according to the fourth embodiment;

[0025] FIG. **16** is a circuit diagram illustrating circuits mounted in the ejection element substrate according to the fifth embodiment;

[0026] FIG. **17** illustrates a table illustrating a relationship between selection control data to be inputted into two selection circuits and heaters to be selected by these selection circuits in the ejection element substrate according to the fifth embodiment;

[0027] FIG. **18** is a circuit diagram illustrating circuits mounted in the ejection element substrate according to the sixth embodiment;

[0028] FIG. **19** is a plan diagram illustrating an entire ejection element substrate according to other embodiments;

[0029] FIG. **20** is a partially-enlarged plan diagram illustrating part of the ejection element substrate according to the other embodiments; and

[0030] FIG. **21** is a partially-enlarged plan diagram illustrating part of the ejection element substrate according to the other embodiments.

DESCRIPTION OF THE EMBODIMENTS

[0031] Embodiments are explained below in detail with reference to the attached drawings. Note that the following embodiments do not limit the disclosure according to the scope of claims. Although multiple characteristics are described in the embodiments, not all of the multiple characteristics are necessarily essential for the disclosure, and the multiple characteristics may be used in any combination. Moreover, in the attached drawings, the same or similar configurations are denoted by the same reference numerals, and overlapping explanation is omitted in some cases.

First Embodiment

[0032] FIG. **1** is a plan diagram illustrating an entire ejection element substrate **100** according to a first embodiment. With reference to FIG. **1**, a heater array **101***a* in which multiple heaters **101** are aligned in one array in a Y direction (also referred to as “first direction” or “main direction”) is arranged in a center portion of the ejection element substrate **100**. A first ink supply port array **102***a* in which multiple ink supply ports **102** are aligned in one array in the Y direction, a first drive circuit array **103***a* in which multiple drive circuits **103** are aligned in one array in the Y direction, and a first selection circuit **105** are arranged on the left side of the heater array **101***a* in this order from the side closer to the heater array **101***a*. A second ink supply port array **102***b* in which multiple ink supply ports **102** are aligned in one array in the Y direction, a second drive circuit array **103***b* in which multiple drive circuits **103** are aligned in one array in the Y direction, and a second selection circuit **106** are arranged on the right side of the heater array **101***a* in this order from the side closer to the heater array **101***a*. A set of the first ink supply port array **102***a*, the first drive circuit array **103***a*, and the first selection circuit **105** is arranged on the opposite side of the heater array **101***a* to the side on which a set of the second ink supply port array **102***b*, the second drive circuit array **103***b*, and the second selection circuit **106** are arranged. Moreover, the set of the first ink supply port array **102***a*, the first drive circuit array **103***a*, and the first selection circuit **105** is arranged to be line symmetric to the set of the second ink supply port array **102***b*, the second drive circuit array **103***b*, and the second selection circuit **106**, with respect to the heater array **101***a*. Note that a terminal array **104** in which multiple terminals are aligned is arranged near each of an upper edge and a lower edge of the ejection element substrate **100**.

[0033] In the heater array **101***a*, the multiple heaters **101** are arranged at a pitch of 1,200 dpi. The heaters **101** are one type of energy generation elements that generate energy and provide the energy to ink being liquid to eject the ink from pressure chambers **156** (see FIG. **2**C) to the outside via ejection ports **500** (see FIGS. **2**A and **2**C). The heaters **101** are arranged at positions where the heaters **101** face the ejection ports **500** across the pressure chambers **156**.

Accordingly, as illustrated in FIG. **2**A, the ejection ports **500** and the heaters **101** are arranged to overlap one another in a plane direction (XY direction) of the ejection element substrate **100**. Moreover, the ejection ports **500** and the heaters **101** are arranged near the centers of the pressure chambers **156**. The energy generation elements are not limited to the heaters, and for example, piezoelectric elements may be used as the energy generation elements.

[0034] Explanation is given below with the respective heaters **101** named as Seg0, Seg1, . . . , Seg62, and Seg63 from the top as illustrated in FIG. **2**A.

[0035] The ink supply ports **102** penetrate the ejection element substrate **100** in a thickness direction (Z direction). Common flow paths **152** and **153** (see FIGS. **2**B and **2**C) communicating with the multiple ink supply ports **102** aligned in the Y direction are formed in the Y direction in a boundary region between the ejection element substrate **100** and an orifice plate **151** (see FIG. **2**C). Moreover, the pressure chambers **156** are also formed in the boundary region, and individual flow paths **154** and **155** (see FIG. **2**C) that allow the individual pressure chambers **156** and the common flow paths **152** and **153** to communicate with one another are also formed in the boundary region. For example, the multiple ink supply ports **102** aligned in the ink supply port array **102***a*, the common flow path **152** communicating with the multiple ink supply ports **102**, and the individual flow paths **154** communicating with the common flow path **152** are used to supply the ink to the pressure chambers **156**. Moreover, the multiple ink supply ports **102** aligned in the ink supply port array **102***b*, the common flow path **153** communicating with the multiple ink supply ports **102**, and the individual flow paths **155** communicating with the common flow path **153** are used to supply the ink to the pressure chambers **156** or to collect the ink from the pressure chambers **156**.

[0036] The heaters **101** are connected to the drive circuits **103** via wiring lines **107** arranged between each adjacent two of the ink supply ports **102**. The multiple drive circuits **103** belonging to the first drive circuit array **103***a* are connected, respectively, to the heaters seg0, Seg2, . . . , and Seg 62 which are even-numbered heaters **101**. The multiple drive circuits **103** belonging to the second drive circuit array **103***b* are connected, respectively, to the heaters seg1, Seg3, . . . , and Seg 63 which are odd-numbered heaters **101**.

[0037] As described later, the first selection circuit **105** selects some of the heaters Seg0, Seg2, . . . , and Seg62 according to, for example, first selection control data obtained by parallelizing first serial data (see FIG. **3**). Then, the first selection circuit **105** causes the drive circuits **103** corresponding to the selected heaters to drive the selected heaters. Accordingly, some of the heaters selected from among the heaters Seg0, Seg2, . . . , and Seg62 according to the first selection control data are driven, and the ink is ejected from the ejection ports **500** corresponding to the selected heaters according to this drive. Note that the first selection circuit **105** directly selects the drive circuits **103** corresponding to the selected heaters **101**. Then, the selected drive circuits **103** drive the corresponding heaters **101** at an intensity depending on a not-illustrated signal. Note that, in the case where the description simply states that the drive circuits **103** drive the heaters **101** below, this means that the drive circuits **103** drive the heaters **101** at the intensity depending on the not-illustrated signal. However, the present disclosure is not limited to this, and the above-mentioned description may also mean that the drive circuits **103** drive the heaters **101** at a fixed intensity.

[0038] As described later, the second selection circuit **106** selects some of the heaters Seg1, Seg3, . . . , and Seg63 according to, for example, second selection control data obtained by parallelizing second serial data (see FIG. **3**). Then, the second selection circuit **106** causes the drive circuits **103** corresponding to the selected heaters to drive the selected heaters. Accordingly, some of the heaters selected from among the heaters Seg1, Seg3, . . . , and Seg63 according to the second selection control data are driven, and the ink is ejected from the ejection ports **500** corresponding to the selected heaters according to this drive. Note that the second selection circuit **106** directly selects the drive circuits **103** corresponding to the selected heaters **101**. Then, the selected drive circuits **103** drive the corresponding heaters **101**.

[0039] In the following explanation, an expression "the selection circuit selects the heaters **101**" also means that the selection circuit selects the drive circuits **103** corresponding to the selected heaters **101**. Moreover, an expression "the selection circuit selects the drive circuits **103**" also means that the selection circuit selects the heaters **101** corresponding to the selected drive circuits **103**. The "selection circuit" herein is the selection circuit **105** or the selection circuit **106**.

[0040] A power supply terminal, a ground terminal, a first serial data terminal **109** (see FIG. **3**) for inputting the first serial data, a second serial data terminal **110** (see FIG. **3**) for inputting the second serial data, and a clock terminal **112** (see FIG. **3**) for inputting a clock (also referred to as "CLK") are arranged side by side in the terminal arrays **104**. Moreover, a latch signal terminal **111** (see FIG. **3**) for inputting a latch signal (also referred to as "LS") and a heater enable terminal **113** for inputting a heater enable signal (also referred to as "_HE") are also arranged side by side in the terminal array **104**. In this case, "_" included in "HE signal" indicates a negative logic.

[0041] FIG. **2**A is a plan diagram in which part of the ejection element substrate **100** according to the first embodiment is enlarged. A surface of the ejection element substrate **100** is covered with the orifice plate **151** in which the ejection ports **500** are formed. As described above, the pressure chambers **156**, the individual flow paths **154** and **155**, and the common flow paths **152** and **153** are also formed in the boundary portion between the ejection element substrate **100** and the orifice plate **151**. In FIG. **2**A, the orifice plate **151** is illustrated in a transparent manner to illustrate the positions of the respective elements in the ejection element substrate **100**. The ejection ports **500** are ports that allow the pressure chambers **156** and the outside to communicate with one another as described above, and penetrate the orifice plate **151**.

[0042] Although this explanation overlaps the explanation made with reference to FIG. **1**, as illustrated in FIG. **2**A, the first ink supply port array **102***a* and the second ink supply port array **102***b* are arranged to be left-right symmetric across the heater array **101***a*. In this case, the heaters **101** are aligned in the heater array **101***a*, and the ejection ports **500** are also similarly aligned. Moreover, the multiple ink supply ports **102** are aligned in the first ink supply port array **102***a*, and the multiple ink supply ports are also aligned in the second ink supply port array **102***b*. Furthermore, the two common flow paths **152** and **153** are arranged at line symmetric positions with respect to the heater array **101***a*.

Moreover, the individual flow paths **154** that allow the one common flow path **152** to communicate with the pressure chambers **156** and the individual flow paths **155** that allow the other common flow path **153** to communicate with the pressure chambers **156** are arranged at line symmetric positions with respect to the heater array **101***a*. The one individual flow path **154** communicating with each pressure chamber **156** allows the pressure chamber **156** to communicate with the common flow path **152** communicating with the multiple ink supply ports **102** in the ink supply port array **102***a*. The other individual flow path **155** communicating with each pressure chamber **156** allows the pressure chamber **156** to communicate with the common flow path **153** communicating with the multiple ink supply ports **102** in the ink supply port array **102***b*. Accordingly, a liquid resistance from the common flow path **152** corresponding to the ink supply port array **102***a* to the pressure chambers **156** and a liquid resistance from the common flow path **153** corresponding to the ink supply port array **102***b* to the pressure chambers **156** are substantially the same.

[0043] As illustrated in FIG. **2**A, a partition **501** is formed between each adjacent two of the heaters **101** in the plane direction (XY direction) of the ejection element substrate **100**. The partition **501** is formed integrally with the orifice plate **151**, and is bonded to the surface of the ejection element substrate **100**.

[0044] Moreover, an array of columnar filters **502** is arranged between the first ink supply port array **102***a* and an array of the heaters **101** and the partitions **501**. Similarly, an array of columnar filters **502** is arranged also between the second ink supply port array **102***b* and the array of heaters **101** and the partitions **501**. Each of the filters **502** is formed integrally with the orifice plate **151**, and is bonded to the surface of the ejection element substrate **100**. Note that, in the example of FIGS. **2**A to **2**C, each pressure chamber **156** is surrounded on six sides by the orifice plate **151**, the heater **101**, two partitions **501**, and two pairs of filters **502**.

[0045] FIG. **3** is a circuit diagram illustrating circuits mounted in the ejection element substrate **100** according to the first embodiment. FIG. **3** particularly illustrates details of the first selection circuit **105** and the second selection circuit **106**.

[0046] The terminal arrays **104** include the clock terminal **112**, the first serial data terminal **109**, and the second serial data terminal **110**. Moreover, the terminal arrays **104** further include the latch signal terminal **111** and the heater enable terminal **113**.

[0047] The clock CLK, the latch signal LS, and the heater enable signal_HE are supplied to both of the first selection circuit **105** and the second selection circuit **106**. However, the first serial data is supplied only to the first selection circuit **105**, and the second serial data is supplied only to the second selection circuit **106**.

[0048] The first selection circuit **105** includes D-type flip-flops (hereinafter, referred to as "DFF") **200** to **203**, **209**, and **210** and latches (hereinafter, referred to as "LT") **204** to **207**, **211**, and **212**. Moreover, the first selection circuit **105** also includes a decoder **208**, logical product operation circuits (hereinafter, referred to as "AND") **213** and **214**, and selectors **215** and **216**. The DFFs **200** to **203**, **209**, and **210** and the LTs **204** to **207**, **211**, and **212** form a conversion circuit configured to convert the first serial data to parallel data.

[0049] The second selection circuit **106** includes DFFs **220** to **223**, **229**, and **230**, LTs **224** to **227**, **231**, and **232**, a decoder **228**, ANDs **233** and **234**, and selectors **235** and **236**. The DFFs **220** to **223**, **229**, and **230** and the LTs **224** to **227**, **231**, and **232** form a conversion circuit configured to convert the second serial data to parallel data.

[0050] The decoder **208** expands the first selection control data encoded in 4-bit binary to 16 BLE signals. The decoder **228** also similarly expands the second selection control data encoded in 4-bit binary to other 16 BLE signals. The BLE signals are signals that individually designate the heater **101** to be selected by the enabled selectors **215**, **216**, **235**, or **236**. Accordingly, each of the selectors **215**, **216**, **235**, and **236** selects and activates the drive circuit **103** for diving the heater **101** corresponding to the BLE signal that is at a HIGH level in the case where the selector is enabled. The selected and activated drive circuit **103** drive the corresponding heater **101**. Note that only one BLE signal is set to the HIGH level among the 16 BLE signals inputted into one selector. The same applies to the embodiments described later.

[0051] In the explanation of the present embodiment, an expression "the selector selects the heater **101**" also means that the selector selects the drive circuit **103** corresponding to the selected heater **101**. Moreover, an expression "the selector selects the drive circuit **103**" also means that the selector selects the heater **101** selected by the selected drive circuit **103**. The "selector" herein is any of the selectors **215**, **216**, **235**, and **236**. The same applies to the embodiments described later.

[0052] The first serial data is taken into the DFF **200** in synchronization with a rising edge of the CLK. Thereafter, the first serial data taken into the DFF **200** is transferred to the DFF **202** and the DFF **209** in sequence in synchronization with the rising edge of the CLK. Moreover, the first serial data is taken into the DFF **201** in synchronization with a falling edge of the CLK. Thereafter, the first serial data taken into the DFF **201** is transferred to the DFF **203** and the DFF **210** in sequence in synchronization with the falling edge of the CLK.

[0053] The LTs **204** to **207**, **211**, and **212** latch Q outputs of the DFFs **200** to **203**, **209**, and **210**, respectively, at a rising edge of the LS.

[0054] Q outputs of the LTs **204** to **207** are supplied to the decoder **208** as the selection control data encoded in binary. The decoder **208** sets one BLE signal designated by the Q outputs of the LTs **204** to **207** among the 16 BLE signals, to HIGH.

[0055] In the case where the _HE goes to Low, outputs of the ANDs **213** and **214** go to High in response to High Q outputs of the LTs **211** and **212**. In the case where the output of the AND **213** is High, the selector **215** selects one of 16 drive circuits **103** depending on the output of the decoder **208**. One of the 16 heaters Seg0, Seg2, . . . , Seg28, and Seg30 is thereby driven by the selected drive circuit **103**. Similarly, in the case where the output of the AND **214** is High, the selector **216** selects one of 16 drive circuits **103** depending on the output of the decoder **208**. One of the 16 heaters Seg32, Seg34, Seg60, and Seg62 is thereby driven by the selected drive circuit **103**.

[0056] The second serial data is taken into the DFF **220** in synchronization with the rising edge of the CLK. Thereafter, the second serial data taken into the DFF **220** is transferred to the DFF **222** and the DFF **229** in sequence in synchronization with the rising edge of the CLK. Moreover, the second serial data is taken into the DFF **221** in synchronization with the falling edge of the CLK. Thereafter, the second serial data taken into the DFF **221** is transferred to the DFF **223** and the DFF **230** in sequence in synchronization with the falling edge of the CLK.

[0057] The LTs **224** to **227**, **231**, and **232** latch Q outputs of the DFFs **220** to **223**, **229**, and **230** at the rising edge of the LS.

[0058] The Q outputs of the LTs **224** to **227** are supplied to the decoder **228** as the selection control data encoded in binary. The decoder **228** sets one BLE signal designated by the Q outputs of the LTs **224** to **227** among the 16 BLE signals to HIGH.

[0059] In the case where the _HE goes to Low, outputs of the ANDs **233** and **234** go to High in response to High Q outputs of the LTs **231** and **232**. In the case where the output of the AND **233** is High, the selector **235** selects one of 16 drive circuits **103** depending on the output of the decoder **228**. One of the 16 heaters Seg1, Seg3, . . . , Seg29, and Seg31 is thereby driven by the selected drive circuit **103**. Similarly, in the case where the output of the AND **234** is High, the selector **236** selects one of 16 drive circuits **103** depending on the output of the decoder **228**. One of the 16 heaters Seg33, Seg35, . . . , Seg61, and Seg63 is thereby driven by the selected drive circuit **103**.

[0060] FIG. **4** is a timing chart illustrating an operation of the circuits mounted in the ejection element substrate **100** according to the first embodiment.

[0061] The first serial data includes D0_EV, D1_EV, BE0_EV, BE1_EV, BE2_EV, and BE3 EV. The D0 EV, the BE0_EV, and the BE2_EV correspond to the rising edge of the CLK. The D1 EV, the BE1_EV, and the BE3_EV correspond to the falling edge of the CLK.

[0062] Moreover, the second serial data includes D0_OD, D1_OD, BE0_OD, BE1_OD, BE2_OD, and BE3_OD. The D0_OD, the BE0_OD, and the BE2_OD correspond to the rising edge of the CLK. The D1_OD, the BE1_OD, and the BE3_OD correspond to the falling edge of the CLK.

[0063] In a period s**1000**, the first serial data and the second serial data for selecting the heaters **101** to be driven in a period s**1001** are inputted in synchronization with the CLK, and are latched at the rising of the LS. In the case where the _HE goes to Low in the period s**1001**, the drive circuits **103** corresponding to the selected heaters are activated, and a heater current thereby flows in the selected heaters.

[0064] Similarly, in the period s**1001**, the first serial data and the second serial data for selecting the heaters **101** to be driven in a period subsequent to the period s**1001** are inputted in synchronization with the CLK, and are latched at the rising of the LS. In the case where the _HE goes to Low in the period subsequent to the period S**1001**, the heater current flows in the selected heaters. This operation is repeated from this point on.

[0065] FIG. **5** illustrates a table illustrating a relationship between the selection control data inputted into the selection circuits **105** and **106** illustrated in FIG. **3** and the heaters **101** selected by these selection circuits **105** and **106**. In this case, the selection control data is included in both of the first serial data and the second serial data. The BE0 EV, the BE1_EV, the BE2_EV, and the BE3_EV included in the first serial data form the first selection control data encoded in binary. The D0_EV, and the D1_EV included in the first serial data are the pieces of enable data for enabling the selectors **215** and **216**, respectively. The BE0_OD, the BE1_OD, the BE2_OD, and the BE3_OD included in the second serial data form the second selection control data encoded in binary. The D0_OD and the D1_OD included in the second serial data are the pieces of enable data for enabling the selectors **235** and **236**, respectively.

[0066] For example, in the case where only one heater Seg20 is to be selected, the values illustrated in FIG. **5** are set in the first serial data and the second serial data. Specifically, the pieces of data included in the first serial data are set as follows.

[0067] D0_EV: High
[0068] D1 EV: Low
[0069] BE0 EV: Low
[0070] BE1_EV: High
[0071] BE2 EV: Low
[0072] BE3_EV: High

[0073] Moreover, the pieces of data included in the second serial data are set as follows.

[0074] D0_OD: Low
[0075] D1_OD: Low
[0076] BE0 to 3_OD: High or Low

[0077] Then, in the case where the _HE goes to Low after the rising of the LS, one drive circuit **103** corresponding to the heater Seg20 is activated, and the heater Seg20 is driven.

[0078] Moreover, for example, in the case where only two heaters Seg20 and Seg 52 are to be selected, the values illustrated in FIG. **5** are set in the first serial data and the second serial data. Specifically, the pieces of data included in the first serial data are set as follows.

[0079] D0_EV: High
[0080] D1_EV: High
[0081] BE0 EV: Low
[0082] BE1_EV: High
[0083] BE2_EV: Low
[0084] BE3_EV: High

[0085] Moreover, the pieces of data included in the second serial data are set as follows.

[0086] D0_OD: Low
[0087] D1_OD: Low
[0088] BE0 to 3_OD: High or Low

[0089] Then, in the case where the HE goes to Low after the rising of the LS, total of two drive circuits **103** corresponding to the heaters Seg20 and Seg 52 are activated, and the heaters Seg20 and Seg 52 are driven.

[0090] Moreover, for example, in the case where four heaters Seg5, Seg20, Seg37, and Seg 52 are to be selected, the values illustrated in FIG. **5** are set in the first serial data and the second serial data. Specifically, the pieces of data included in the first serial data are set as follows.

[0091] D0_EV: High
[0092] D1_EV: High
[0093] BE0 EV: Low
[0094] BE1_EV: High
[0095] BE2 EV: Low
[0096] BE3_EV: High

[0097] Moreover, the pieces of data included in the second serial data are set as follows.

[0098] D0_OD: High
[0099] D1_OD: High
[0100] BE0 OD: Low
[0101] BE1_OD: High
[0102] BE2 OD: Low
[0103] BE3 OD: Low

[0104] Then, in the case where the _HE goes to Low after the rising of the LS, total of four drive circuits **103** corresponding to the heaters Seg5, Seg20, Seg37, and Seg 52 are activated, and the heaters Seg5, Seg20, Seg37, and Seg 52 are driven.

[0105] Zero to four drive circuits **103** can be activated depending on the number of pieces of data set to High among the D0_EV, the D1_EV, the D0_OD, and the D1_OD, and zero to four heaters can be driven in response to this activation.

[0106] In this case, the selectors **215**, **216**, **235**, and **236** are enabled in the case where the D0_EV, the D1_EV, the D0_OD, and the D1_OD go to High, respectively. Then, in the case where the selectors **215** and **216** are enabled, the selectors **215** and **216** each select one heater **101** designated by the BE0_EV, the BE1_EV, the BE2_EV, and the BE3_EV, and cause the drive circuit **103** corresponding to the selected heater to drive the heater. Similarly, in the case where the selectors **235** and **236** are enabled, the selectors **235** and **236** each select one heater **101** designated by the BE0_OD, the BE1_OD, the BE2_OD, and the BE3_OD, and cause the drive circuit **103** corresponding to the selected heater to drive the heater. Accordingly, in the case where the circuit illustrated in FIG. **3** is viewed as a whole, a maximum of four heaters **101** are simultaneously driven by the drive circuits **103** corresponding to the respective heaters **101**. In a view limited to the first selection circuit **105**, a maximum of two heaters **101** are simultaneously driven by the drive circuits **103** corresponding to the respective heaters **101**. Similarly, in a view limited to the second selection circuit **106**, a maximum of two heaters **101** are simultaneously driven by the drive circuits **103** corresponding to the respective heaters **101**. Accordingly, the first selection circuit **105** includes as many selectors as the maximum total number of heaters **101** simultaneously selectable by the first selection circuit **105**. Similarly, the second selection circuit **106** includes as many selectors as the maximum total number of heaters **101** simultaneously selectable by the second selection circuit **106**.

[0107] Since the heaters arranged in one array at high density can be driven as described above, the ejection element substrate **100** downsized, and the liquid ejection head including the ejection element substrate **100** can be also downsized.

Second Embodiment

[0108] An ejection element substrate **100** according to a second embodiment has a basic configuration as illustrated in FIG. **1**, like the ejection element substrate **100** according to the first embodiment.

[0109] FIG. **6** is a circuit diagram illustrating circuits mounted in the ejection element substrate **100** according to the second embodiment. FIG. **6** particularly illustrates details of the first selection circuit **105** and the second selection circuit **106**.

[0110] The terminal array **104** includes the clock terminal **112**, a serial data terminal **115** for inputting serial data, the latch signal terminal **111**, and the heater enable terminal **113**.

[0111] The clock CLK, the latch signal LS, and the heater enable signal_HE are supplied to both of the first selection circuit **105** and the second selection circuit **106**. The serial data is also supplied to both of the first selection circuit **105** and the second selection circuit **106**.

[0112] The first selection circuit **105** includes DFFs **300** to **303**, **309**, and **310**, LTs **304** to **307**, **311**, and **312**, the decoder **208**, ANDs **313** and **314**, and the selectors **215** and **216**. The second selection circuit **106** includes DFFs **320** to **323**, **329**, and **330**, LTs **324** to **327**, **331**, and **332**, the decoder **228**, ANDs **333** and **334**, and the selectors **235** and **236**. The DFFs **300** to **303**, **309**, and **310** and the LTs **304** to **307**, **311**, and **312** form a conversion circuit configured to convert a part of the serial data to parallel data. The DFFs **320** to **323**, **329**, and **330** and the LTs **324** to **327**, **331**, and **332** form a conversion circuit configured to convert another part of the serial data to parallel data.

[0113] The serial data is taken into the DFF **300** in synchronization with the rising edge of the CLK. Thereafter, the serial data taken into the DFF **300** is transferred to the DFF **301**, the DFF **302**, the DFF **303**, the DFF **309**, and the DFF **310** in sequence in synchronization with the rising edge of the CLK.

[0114] Moreover, the serial data is taken into the DFF **320** in synchronization with the falling edge of the CLK. Thereafter, the serial data taken into the DFF **320** is transferred to the DFF **321**, the DFF **322**, the DFF **323**, the DFF **329**, and the DFF **330** in sequence in synchronization with the falling edge of the CLK.

[0115] The LTs **304** to **307**, **311**, **312**, **324** to **327**, **331**, and **332** latch Q outputs of the DFFs **300** to **303**, **309**, **310**, **320** to **323**, **329**, and **330**, respectively, at the rising edge of the LS.

[0116] Q outputs of the LTs **304** to **307** are supplied to the decoder **208** as selection signals encoded in binary. The decoder **208** sets one BLE signal designated by the Q outputs of the LTs **304** to **307** among 16 BLE signals, to HIGH.

[0117] Similarly, Q outputs of the LTs **324** to **327** are supplied to the decoder **228** as selection signals encoded in binary. The decoder **228** sets one BLE signal designated by the Q outputs of the LTs **324** to **327** among 16 BLE signals, to HIGH.

[0118] In the case where the _HE goes to Low, outputs of the ANDs **313** and **314** go to High in response to High Q outputs of the LTs **311** and **312**. In the case where the output of the AND **314** is High, the selector **215** selects one of 16 drive circuits **103** depending on the output of the decoder **208**. One of the 16 heaters Seg0, Seg2, . . . , Seg28, and Seg30 is thereby driven by the selected drive circuit **103**. Similarly, in the case where the output of the AND **313** is High, the selector **216** selects one of 16 drive circuits **103** depending on the output of the decoder **208**. One of the 16 heaters Seg32, Seg34, . . . Seg60, and Seg62 is thereby driven by the selected drive circuit **103**.

[0119] Moreover, in the case where the _HE goes to Low, outputs of the ANDs **333** and **334** go to High in response to High Q outputs of the LTs **331** and **332**. In the case where the output of the AND **334** is High, the selector **235** selects one of 16 drive circuits **103** depending on the output of the decoder **228**. One of the 16 heaters Seg1, Seg3, . . . , Seg29, and Seg31 is thereby driven by the selected drive circuit **103**. Similarly, in the case where the output of the AND **333** is High, the selector **236** selects one of 16 drive circuits **103** depending on the output of the decoder **228**. One of the 16 heaters Seg33, Seg35, . . . , Seg61, and Seg63 is thereby driven by the selected drive circuit **103**.

[0120] FIG. **7** is a timing chart illustrating an operation of the circuits mounted in the ejection element substrate **100** according to the second embodiment.

[0121] The serial data includes the D0_EV, the D0_OD, the D1_EV, the D1_OD, the BE0_EV, the BE0_OD, the BE1_EV, the BE1_OD, the BE2_EV, the BE2_OD, the BE3_EV, and the BE3_OD. The D0_EV, the D1_EV, the BE0_EV, the BE1 EV, the BE2_EV, and the BE3_EV correspond to the rising edge of the CLK. The D0_OD, the D1_OD, the BE0_OD, the BE1_OD, the BE2_OD, and the BE3_OD correspond to the falling edge of the CLK.

[0122] In a period s**1100**, the serial data for selecting the heaters **101** to be driven in a period s**1101** is inputted in synchronization with the CLK, and is latched at the rising of the LS. In the case where the HE goes to Low in the period s**1101**, the heater current flows in the selected heaters.

[0123] Similarly, in the period s**1101**, the serial data for selecting the heaters **101** to be driven in a period subsequent to the period s**1101** is inputted in synchronization with the CLK, and is latched at the rising of the LS. In the case where the _HE goes to Low in the period subsequent to the period S**1101**, the heater current flows in the selected heaters. This operation is repeated from this point on.

[0124] FIG. **8** illustrates a table illustrating a relationship between the selection control data inputted into the selection circuits **105** and **106** illustrated in FIG. **6** and the heaters **101** selected by these selection circuits **105** and **106**. In this case, the selection control data is included in the serial data. The BE0 EV, the BE1_EV, the BE2_EV, and the BE3 EV included in the serial data form the first selection control data encoded in binary. The D0 EV and the D1 EV included in the serial data are the pieces of enable data for enabling the selectors **215** and **216**, respectively. The BE0_OD, the BE1_OD, the BE2_OD, and the BE3_OD included in the serial data form the second selection control data encoded in binary. The D0_OD and the D1_OD included in the serial data are the pieces of enable data for enabling the selectors **235** and **236**, respectively.

[0125] For example, in the case where only one heater Seg20 is to be selected, the values illustrated in FIG. **8** are set in the serial data. Specifically, the pieces of data included in the serial data are set as follows.

[0126] D0_EV: High

[0127] D1 EV: Low

[0128] BE0 EV: Low

[0129] BE1_EV: High

[0130] BE2 EV: Low

[0131] BE3_EV: High

[0132] Moreover, the other pieces of data included in the serial data are set as follows.

[0133] D0 OD: Low

[0134] D1_OD: Low

[0135] BE0 to 3_OD: High or Low

[0136] Then, in the case where the _HE goes to Low after the rising of the LS, one drive circuit **103** corresponding to the heater Seg20 is activated, and the heater Seg20 is driven.

[0137] Moreover, for example, in the case where only two heaters Seg20 and Seg 52 are to be selected, the values illustrated in FIG. **8** are set in the serial data. Specifically, the pieces of data included in the serial data are set as follows.

[0138] D0_EV: High

[0139] D1_EV: High

[0140] BE0_EV: Low

[0141] BE1_EV: High

[0142] BE2 EV: Low

[0143] BE3_EV: High

[0144] Moreover, the other pieces of data included in the serial data are set as follows.

[0145] D0_OD: Low

[0146] D1_OD: Low

[0147] BE0 to 3_OD: High or Low

[0148] Then, in the case where the _HE goes to Low after the rising of the LS, total of two drive circuits **103** corresponding to the heaters Seg20 and Seg 52 are activated, and the heaters Seg20 and Seg 52 are driven.

[0149] Moreover, for example, in the case where four heaters Seg5, Seg20, Seg37, and Seg 52 are to be selected, the values illustrated in FIG. **8** are set in the serial data. Specifically, the pieces of data included in the serial data are set as follows.

[0150] D0_EV: High

[0151] D1_EV: High

[0152] BE0_EV: Low

[0153] BE1_EV: High

[0154] BE2 EV: Low

[0155] BE3_EV: High

[0156] Moreover, the other pieces of data included in the serial data are set as follows.

[0157] D0_OD: High

[0158] D1_OD: High

[0159] BE0 OD: Low

[0160] BE1_OD: High

[0161] BE2 OD: Low

[0162] BE3_OD: Low

[0163] Then, in the case where the _HE goes to Low after the rising of the LS, total of four drive circuits **103** corresponding to the heaters Seg5, Seg20, Seg37, and Seg 52 are activated, and the heaters Seg5, Seg20, Seg37, and Seg 52 are driven.

[0164] Zero to four drive circuits **103** can be activated depending on the number of pieces of data set to High among the D0_EV, the D1_EV, the D0_OD, and the D1_OD, and zero to four heaters can be driven in response to this activation.

[0165] In this case, the selectors **215**, **216**, **235**, and **236** are enabled in the case where the D0_EV, the D1_EV, the D0_OD, and the D1_OD go to High, respectively. Then, in the case where the selectors **215** and **216** are enabled, the selectors **215** and **216** each select one heater **101** designated by the BE0_EV, the BE1_EV, the BE2_EV, and the BE3 EV, and cause the drive circuit **103** corresponding to the selected heater to drive the heater. Similarly, in the case where the selectors **235** and **236** are enabled, the selectors **235** and **236** each select one heater **101** designated by the BE0_OD, the BE1_OD, the BE2_OD, and the BE3_OD, and cause the drive circuit **103** corresponding to the selected heater to drive the heater. Accordingly, in the case where the circuit illustrated in FIG. **6** is viewed as a whole, a maximum of four heaters **101** are simultaneously driven by the drive circuits **103** corresponding to the respective heaters **101**. In a view limited to the first selection circuit **105**, a maximum of two heaters **101** are simultaneously driven by the drive circuits **103** corresponding to the respective heaters **101**. Similarly, in a view limited to the second selection circuit **106**, a maximum of two heaters **101** are simultaneously driven by the drive circuits **103** corresponding to the respective heaters **101**. Accordingly, the first selection circuit **105** includes as many selectors as the maximum total number of heaters **101** simultaneously selectable by the first selection circuit **105**. Similarly, the second selection circuit **106** includes as many selectors as the maximum total number of heaters **101** simultaneously selectable by the second selection circuit **106**.

[0166] Since the heaters arranged in one array at high density can be driven as described above, the ejection element substrate **100** can be downsized, and the liquid ejection head including the ejection element substrate **100** can be also downsized. Moreover, since the number of electrodes for data input can be reduced, it is possible to reduce sealing regions for preventing invasion of the ink into the electrodes and improve reliability.

Third Embodiment

[0167] An ejection element substrate **100** according to the third embodiment has a basic configuration as illustrated in FIG. **1**, like the ejection element substrate **100** according to the first embodiment.

[0168] FIG. **9** is a circuit diagram illustrating circuits mounted in the ejection element substrate **100** according to the third embodiment. FIG. **6** particularly illustrates details of the first selection circuit **105** and the second selection circuit **106**.

[0169] The terminal array **104** includes the clock terminal **112**, a serial data terminal **116** for inputting serial data, the latch signal terminal **111**, and the heater enable terminal **113**.

[0170] The clock CLK, the latch signal LS, and the heater enable signal_HE are supplied to both of the first selection circuit **105** and the second selection circuit **106**. The serial data is also supplied to both of the first selection circuit **105** and the second selection circuit **106**.

[0171] The first selection circuit **105** includes DFFs **400** to **403**, **409**, and **410**, LTs **404** to **407**, **411**, and **412**, the decoder **208**, ANDs **413**, **414**, **417**, and **418**, and selectors **415**, **416**, **419**, and **420**. The second selection circuit **106** includes DFFs **400** to **403**, **421**, and **422**, LTs **404** to **407**, **423**, and **424**, the decoder **208**, ANDs **425**, **426**, **429**, and **430**, and selectors **427**, **428**, **431**, and **432**. Note that the DFFs **400** to **403**, the LTs **404** to **407**, and the decoder **208** are shared by the first selection circuit **105** and the second selection circuit **106**. The DFFs **400** to **403**, **409**, **410**, **421** and **422** and the LTs **404** to **407**, **411**, **412**, **423**, and **424** form a conversion circuit configured to convert the serial data to parallel data.

[0172] The serial data is taken into the DFF **400** in synchronization with the rising edge of the CLK. Thereafter, the serial data taken into the DFF **400** is transferred to the DFF **402**, the DFF **409**, and the DFF **410** in sequence in synchronization with the rising edge of the CLK.

[0173] Moreover, the serial data is taken into the DFF **401** in synchronization with the falling edge of the CLK. Thereafter, the serial data taken into the DFF **401** is transferred to the DFF **403**, the DFF **421**, and the DFF **422** in sequence in synchronization with the falling edge of the CLK.

[0174] The LTs **404** to **407**, **411**, **412**, **423**, and **424** latch Q outputs of the DFFs **400** to **403**, **409**, **410**, **421**, and **422**, respectively, at the rising edge of the LS.

[0175] The Q outputs of the LTs **404** to **407** are supplied to the decoder **208** as the selection signals encoded in binary. The decoder **208** sets one BLE signal designated by the Q outputs of the LTs **404** to **407** among 16 BLE signals to HIGH.

[0176] In the case where the _HE goes to Low, outputs of the ANDs **413** and **429** goes to High in response to a High Q output of the LT **411**. Moreover, in the case where the HE goes to Low, outputs of the ANDs **414** and **430** go to High in response to a High Q output of the LT **412**. Furthermore, in the case where the _HE goes to Low, outputs of the ANDs **417** and **425** go to High in response to a High Q output of the LT **423**. Moreover, in the case where the _HE goes to Low, outputs of the ANDs **418** and **426** go to High in response to a High Q output of the LT **424**.

[0177] In this case, eight out of the 16 BLE signals outputted by the decoder **208** are supplied to the selectors **416**, **420**, **415**, and **419**, and the other eight BLE signals are supplied to the selectors **432**, **428**, **431**, and **427**.

[0178] In the case where the output of the AND **414** is High and one of the eight BLE signals supplied to the selector **416** is High, the selector **416** selects one of eight drive circuits **103** depending on the BLE selection signal that is High. One of the eight heaters Seg0, Seg2, . . . , Seg12, and Seg14 is thereby driven by the selected drive circuit **103**.

[0179] In the case where the output of the AND **418** is High and one of the eight BLE signals supplied to the selector **420** is High, the selector **420** selects one of eight drive circuits **103** depending on the BLE selection signal that is High. One of the eight heaters Seg16, Seg18, . . . , Seg28, and Seg30 is thereby driven by the selected drive circuit **103**.

[0180] In the case where the output of the AND **413** is High and one of the eight BLE signals supplied to the selector **415** is High, the selector **415** selects one of eight drive circuits **103** depending on the BLE selection signal that is High. One of the eight heaters Seg32, Seg34, . . . , Seg44, and Seg46 is thereby driven by the selected drive circuit **103**.

[0181] In the case where the output of the AND **417** is High and one of the eight BLE signals supplied to the selector **419** is High, the selector **419** selects one of eight drive circuits **103** depending on the BLE selection signal that is High. One of the eight heaters Seg48, Seg50, . . . , Seg60, and Seg62 is thereby driven by the selected drive circuit **103**.

[0182] In the case where the output of the AND **430** is High and one of the eight BLE signals supplied to the selector **432** is High, the selector **432** selects one of eight drive circuits **103** depending on the BLE selection signal that is High. One of the eight heaters Seg1, Seg3, . . . , Seg13, and Seg15 is thereby driven by the selected drive circuit **103**.

[0183] In the case where the output of the AND **426** is High and one of the eight BLE signals supplied to the selector **428** is High, the selector **428** selects one of eight drive circuits **103** depending on the BLE selection signal that is High. One of the eight heaters Seg17, Seg19, . . . , Seg29, and Seg31 is thereby driven by the selected drive circuit **103**.

[0184] In the case where the output of the AND **429** is High and one of the eight BLE signals supplied to the selector **431** is High, the selector **431** selects one of eight drive circuits **103** depending on the BLE selection signal that is High. One of the eight heaters Seg33, Seg35, . . . , Seg45, and Seg47 is thereby driven by the selected drive circuit **103**.

[0185] In the case where the output of the AND **425** is High and one of the eight BLE signals supplied to the selector **427** is High, the selector **427** selects one of eight drive circuits **103** depending on the BLE selection signal that is High. One of the eight heaters Se49, Seg51, . . . , Seg61, and Seg63 is thereby driven by the selected drive circuit **103**.

[0186] FIG. **10** is a timing chart illustrating an operation of the circuits mounted in the ejection element substrate **100** according to the third embodiment.

[0187] The serial data includes D0 to D3 and BE0 to BE3. The D0, the D2, the BE0, and the BE2 correspond to the rising edge of the CLK, and the D1, the D3, the BE1, and the BE3 correspond to the falling edge of the CLK.

[0188] In a period s**1200**, the serial data for selecting the heaters **101** to be driven in a period s**1201** is inputted in synchronization with the CLK, and is latched at the rising of the LS. In the case where the HE goes to Low in the period s**1201**, the heater current flows in the selected heaters.

[0189] Similarly, in the period s**1201**, the serial data for selecting the heaters **101** to be driven in a period subsequent to the period s**1201** is inputted in synchronization with the CLK, and is latched at the rising of the LS. In the case where the HE goes to Low in the period subsequent to the period S**1201**, the heater current flows in the selected heaters. This operation is repeated from this point on.

[0190] FIG. **11** illustrates a table illustrating a relationship between the selection control data inputted into the selection circuits **105** and **106** illustrated in FIG. **9** and the heaters **101** selected by these selection circuits **105** and **106**. In this case, the selection control data is included in the serial data. The BE0, the BE1, the BE2, and the BE3 included in the serial data form the selection control data encoded in binary. The D0 included in the serial data is enable data for enabling the selectors **416** and **432**. The D1 included in the serial data is enable data for enabling the selectors **420** and **428**. The D2 included in the serial data is enable data for enabling the selectors **415** and **431**. The D3 included in the serial data is enable data for enabling the selectors **419** and **427**.

[0191] For example, in the case where only one heater Seg20 is to be selected, the values illustrated in FIG. **11** are set in the serial data. Specifically, the pieces of data included in the serial data are set as follows.

[0192] D0: Low

[0193] D1: High

[0194] D2 to 3: Low

[0195] BE0: Low

[0196] BE1: Low

[0197] BE2: High

[0198] BE3: Low

[0199] Then, in the case where the _HE goes to Low after the rising of the LS, one drive circuit **103** corresponding to the heater Seg20 is activated, and the heater Seg20 is driven.

[0200] Moreover, for example, in the case where only two heaters Seg20 and Seg 52 are to be selected, the values illustrated in FIG. **11** are set in the serial data. Specifically, the pieces of data included in the serial data are set as follows.

[0201] D0: Low

[0202] D1: High

[0203] D2: Low

[0204] D3: High

[0205] BE0: Low

[0206] BE1: Low

[0207] BE2: High

[0208] BE3: Low

[0209] Then, in the case where the _HE goes to Low after the rising of the LS, total of two drive circuits **103** corresponding to the heaters Seg20 and Seg 52 are activated, and the heaters Seg20 and Seg 52 are driven.

[0210] Moreover, for example, in the case where four heaters Seg4, Seg20, Seg36, and Seg 52 are to be selected, the values illustrated in FIG. **11** are set in the serial data. Specifically, the pieces of data included in the serial data are set as follows.

[0211] D0 to D3: High

[0212] BE0: Low

[0213] BE1: Low

[0214] BE2: High

[0215] BE3: Low

[0216] Then, in the case where the _HE goes to Low after the rising of the LS, total of four drive circuits **103** corresponding to the heaters Seg4, Seg20, Seg36, and Seg 52 are activated, and the heaters Seg4, Seg20, Seg36, and Seg 52 are driven.

[0217] Zero to four drive circuits **103** can be activated depending on the number of pieces of data set to High among the D0 to D3, and zero to four heaters can be driven in response to this activation.

[0218] In this case, the selectors **416** and **432** are enabled in the case where the D0 goes to High. The selectors **420** and **428** are enabled in the case where the D1 goes to High. The selectors **415** and **431** are enabled in the case where the D2 goes to High. The selectors **419** and **427** are enabled in the case where the D3 goes to High. Then, in the case where the set of the selectors **416** and **432** is enabled, the set of the selectors **416** and **432** selects one heater **101** designated by the BE0, the BE1, the BE2, and the BE3, and causes the drive circuit **103** corresponding to the selected heater to drive the heater. In this situation, there are the case where the selector **416** selects one heater **101** and the selector **432** selects none of the heaters **101** and the case where the selector **416** selects none of the heaters **101** and the selector **432** selects one heater **101**. Similarly, in the case where the set of the selectors **420** and **428** is enabled, the set of the selectors **420** and **428** selects one heater **101** designated by the BE0, the BE1, the BE2, and the BE3, and causes the drive circuit **103** corresponding to the selected heater to drive the heater. In this situation, there are the case where the selector **420** selects one heater **101** and the selector **428** selects none of the heaters **101** and the case where the selector **420** selects none of the heaters **101** and the selector **428** selects one heater **101**. Similarly, in the case where the set of the selectors **415** and **431** is enabled, the set of the selectors **415** and **431** selects one heater **101** designated by the BE0, the BE1, the BE2, and the BE3, and causes the drive circuit **103** corresponding to the selected heater to drive the heater. In this situation, there are the case where the selector **415** selects one heater **101** and the selector **431** selects none of the heaters **101** and the case where the selector **415** selects none of the heaters **101** and the selector **431** selects one heater **101**. Similarly, in the case where the set of the selectors **419** and **427** is enabled, the set of the selectors **419** and **427** selects one heater **101** designated by the BE0, the BE1, the BE2, and the BE3, and causes the drive circuit **103** corresponding to the selected heater to drive the heater. In this situation, there are the case where the selector **419** selects one heater **101** and the selector **427** selects none of the heaters **101** and the case where the selector **419** selects none of the heaters **101** and the selector

427 selects one heater **101**. Accordingly, in the case where the circuit illustrated in FIG. **6** is viewed as a whole, a maximum of four heaters **101** are simultaneously driven by the drive circuits **103** corresponding to the respective heaters **101**. In a view limited to the first selection circuit **105**, a maximum of four heaters **101** are simultaneously driven by the drive circuits **103** corresponding to the respective heaters **101**. Similarly, in a view limited to the second selection circuit **106**, a maximum of four heaters **101** are simultaneously driven by the drive circuits **103** corresponding to the respective heaters **101**. Accordingly, the first selection circuit **105** includes as many selectors as the maximum total number of heaters **101** simultaneously selectable by the set of the first selection circuit **105** and the second selection circuit **106**. Similarly, the second selection circuit **106** includes as many selectors as the maximum total number of heaters **101** simultaneously selectable by the set of the first selection circuit **105** and the second selection circuit **106**.

[0219] Since the heaters arranged in one array at high density can be driven as described above, the ejection element substrate **100** can be downsized, and the liquid ejection head including the ejection element substrate **100** can be also downsized. Moreover, it is possible to reduce the DFFs, the LTs, and the decoders and further downsize the ejection element substrate **100** from those in the first and second embodiments, and the liquid ejection head including the ejection element substrate **100** can be also downsized. Furthermore, since the number of electrodes for data input can be reduced, it is possible to reduce the sealing regions for preventing invasion of the ink into the electrodes and improve reliability, as in the second embodiment.

Fourth Embodiment

[0220] FIG. **12** is a plan diagram illustrating an entire ejection element substrate **100** according to a fourth embodiment.

[0221] The ejection element substrate according to the fourth embodiment is different from the ejection element substrate according to the first to third embodiments illustrated in FIG. **1** in at least the following two points.

[0222] A first difference is as follows. In the ejection element substrate **100** according to the first to third embodiments, as illustrated in FIG. **1**, the ink supply ports **102** included in the first ink supply port array **102***a* and the ink supply ports **102** included in the second ink supply port array **102***b* are arranged in a rectangular lattice pattern of two arrays. Accordingly, each of at least some of the ink supply ports **102** included in the first ink supply port array **102***a* and a corresponding one of at least some of the ink supply ports **102** included in the second ink supply port array **102***b* are arranged at the same position in the direction (Y direction) of the heater array **101***a*. Meanwhile, in the ejection element substrate **100** according to the fourth embodiment, as illustrated in FIG. **12**, the ink supply ports **102** included in the first ink supply port array **102***a* and the ink supply ports **102** included in the second ink supply port array **102***b* are arranged in a staggered pattern of two arrays. Accordingly, at least some of the ink supply ports **102** included in the first ink supply port array **102***a* and at least some of the ink supply ports **102** included in the second ink supply port array **102***b* are arranged alternately in the direction (Y direction) of the heater array **101***a*.

[0223] The other difference is as follows. In the ejection element substrate **100** of the first embodiment, as illustrated in FIG. **1**, every other heater **101** among the multiple heaters **101** included in the heater array **101***a* are connected to the drive circuits **103** included in the first drive circuit array **103***a*, and the other multiple heaters **101** are connected to the drive circuits **103** included in the second drive circuit array **103***b*. As an example, the heaters Seg0, Seg2, . . . , Seg60, and Seg62 are connected to the drive circuits **103** included in the first drive circuit array **103***a*. Moreover, the heaters Seg1, Seg3, . . . , Seg61, and Seg63 are connected to the drive circuits **103** included in the second drive circuit array **103***b*. The same applies to the ejection element substrates **100** according to the second and third embodiments. Meanwhile, in the ejection element substrate **100** according to the fourth embodiment, the multiple heaters **101** included in the heater array **101***a* are connected to the drive circuits **103** as follows. Specifically, as illustrated in FIG. **12**, every other set of four heaters are connected to the drive circuits **103** included in the first drive circuit array **103***a*, and the other sets of four heaters are connected to the drive circuits **103** included in the second drive circuit array **103***b*. As an example, the heaters Seg0 to Seg3, Seg8 to Seg11, . . . , and Seg56 to Seg59 are connected to the drive circuits **103** included in the first drive circuit array **103***a*. Moreover, the heaters Seg4 to Seg7, Seg12 to Seg15, . . . , and Seg60 to Seg63 are connected to the drive circuits **103** included in the second drive circuit array **103***b*. More differences are described later.

[0224] In the ejection element substrate **100** according to the first embodiment, as illustrated in FIG. **1**, one of the four wiring lines **107** arranged in a region between each adjacent two of the ink supply ports **102** has a linear shape, but the other three wiring lines **107** do not have the linear shape. Two of the other three wiring lines **107** each have the following portions between the heaters **101** and the ink supply ports **102**. Specifically, the two (for example, two wiring lines corresponding to the heaters Seg0 and Seg2, respectively) of the other three wiring lines each have a portion extending in a direction (X direction (also referred to as "second direction" or "sub-direction")) orthogonal to the direction (Y direction) of the heater array **101***a* and a portion extending in the same direction (Y direction) as the direction (Y direction) of the heater array **101***a*. Accordingly, a region for arranging at least two wiring lines side by side in the X direction needs to be provided between the heater array **101***a* and each ink supply port array (for example, the ink supply port array **102***a*).

[0225] Meanwhile, in the ejection element substrate **100** according to the fourth embodiment, as illustrated in FIG. **12**, all of the four wiring lines **107** arranged in the region between each adjacent two of the ink supply ports **102** have a linear shape extending in the X direction at least in a predetermined region. In this case, the predetermined region is a region obtained by connecting the region between each adjacent two of the ink supply ports **102** and a region from the region between each adjacent two of the ink supply ports **102** to positions where the heaters **101** are present. Accordingly, in the ejection element substrate **100** according to the fourth embodiment, none of the wiring lines **107** has the portion in which the wiring line extends in the direction (Y direction) of the heater array **101***a*, in the region between the heater array **101***a* and the ink supply port array **102***a*. Hence, there is no need to provide the region for arranging the wiring lines between the heater array **101***a* and the ink supply port array **102***a*. Thus, the ink supply port array **102***a* can be arranged closer to the heater array **101***a*. Similarly, none of the wiring lines **107** has the portion in which the wiring line extends in the direction (Y direction) of the heater array **101***a*, in the region between the heater array **101***a* and the ink supply port array **102***b*. Hence, there is no need to provide the region for arranging the wiring lines between the heater array **101***a* and the ink supply port array **102***b*. Thus, the ink supply port array **102***b* can be arranged closer to the heater array **101***a*. Accordingly, a time period taken to supply the ink from the ink supply ports **102** to the pressure chambers **156** and a time period taken to collect the ink from the pressure chambers **156** into the ink supply ports can be reduced. This enables high-speed printing. Moreover, since there is no need to provide the region for arranging the wiring lines in either the region between the heater array **101***a* and the ink supply port array **102***a* or the region between the heater array **101***a* and the ink supply port array **102***b*, the ejection element substrate **100** can be downsized.

[0226] FIG. **13** is a plan diagram in which part of the ejection element substrate **100** according to the fourth embodiment is enlarged. The ejection element substrate **100** according to the fourth embodiment is different from the configuration illustrated in FIG. **2** in that the ink supply ports **102** in the ink supply port array **102***a* and the ink supply port array **102***b* are arranged in the staggered pattern.

[0227] Since the configuration of the ejection element substrate **100** according to the fourth embodiment and the configuration of the orifice plate **151** attached to this ejection element substrate **100** are basically the same as the configuration of the ejection element substrate **100** according to the first embodiment and the configuration of the orifice plate **151** attached to this ejection element substrate **100**, only the differences are explained. The differences are such that, as explained with reference to FIG. **12**, the multiple ink supply ports **102** belonging to the first ink supply port array **102***a* and the multiple ink supply ports **102** belonging to the second ink supply port array **102***b* are arranged in the staggered pattern of two arrays.

[0228] FIG. **14** is a circuit diagram illustrating circuits mounted in the ejection element substrate **100** according to the fourth embodiment. A basic configuration of the circuits mounted in the ejection element substrate **100** according to the fourth embodiment is the same as the basic configuration of the circuits mounted in the ejection element substrate **100** according to the first embodiment. The connection relationship between the drive circuits **103** and the heaters **101** in the ejection element substrate **100** according to the first embodiment is as illustrated in FIG. **1**, and is also illustrated in FIG. **3**. Meanwhile, a connection relationship between the drive circuits **103** and the heaters **101** in the ejection element substrate **100** according to the fourth embodiment is as illustrated in FIG. **12**, and is also illustrated in FIG. **14**. As apparent from FIGS. **1**, **3**, **12**, and **14**, the ejection element substrate **100** according to the fourth embodiment is different from the ejection element substrate **100** according to the first embodiment only in the connection relationship between the drive circuits **103** and the heaters **101**. Accordingly, only the connection relationship between the drive circuits **103** and the heaters **101** is explained, and overlapping explanation of the common parts is omitted.

[0229] 16 drive circuits **103** selectable by the selector **215** are connected to the heaters Seg0 to 3, Seg8 to 11, Seg16 to 19, and Seg24 to 27, respectively. 16 drive circuits **103** selectable by the selector **216** are connected to the heaters Seg32 to 35, Seg40 to 43, Seg48 to 51, and Seg56 to 59, respectively. 16 drive circuits **103** selectable by the selector **235** are connected to the heaters Seg4 to 7, Seg12 to 15, Seg20 to 23, and Seg28 to 31, respectively. 16 drive circuits **103** selectable by the selector **236** are connected to the heaters Seg36 to 39, Seg44 to 47, Seg52 to 55, and Seg60 to 63, respectively.

[0230] An operation of the circuits mounted in the ejection element substrate **100** according to the fourth embodiment is the same as the operation of the circuits mounted in the ejection element substrate **100** according to the first embodiment. Since this operation is already explained with reference to FIG. **4**, overlapping explanation is omitted.

[0231] FIG. **15** illustrates a table illustrating a relationship between the selection control data inputted into the selection circuits **105** and **106** illustrated in FIG. **14** and the heaters **101** selected by these selection circuits **105** and **106**. In this case, the selection control data is included in both of the first serial data and the second serial data. The BE0_EV, the BE1_EV, the BE2 EV, and the BE3_EV included in the first serial data form the first selection control data encoded in binary. The D0_EV and the D1 EV included in the first serial data are the pieces of enable data for enabling the selectors **215** and **216**, respectively. The BE0_OD, the BE1_OD, the BE2_OD, and the BE3_OD included in the second serial data form the second selection control data encoded in binary. The D0_OD and the D1_OD included in the second serial data are the pieces of enable data for enabling the selectors **235** and **236**, respectively.

[0232] For example, in the case where only one heater Seg20 is to be selected, the values illustrated in FIG. **15** are set in the first serial data and the second serial data. Specifically, the pieces of data included in the first serial data are set as follows.

[0233] D0 EV: Low

[0234] D1 EV: Low

[0235] BE0 to 3_EV: High or Low

[0236] Moreover, the pieces of data included in the second serial data are set as follows.

[0237] D0_OD: High

[0238] D1_OD: Low

[0239] BE0 to 2: Low

[0240] BE3_OD: High

[0241] Then, in the case where the _HE goes to Low after the rising of the LS, one drive circuit **103** corresponding to the heater Seg20 is activated, and the heater Seg20 is driven.

[0242] Moreover, for example, in the case where only two heaters Seg20 and Seg 52 are to be selected, the values illustrated in FIG. **15** are set in the first serial data and the second serial data. Specifically, the pieces of data included in the first serial data are set as follows.

[0243] D0 EV: Low

[0244] D1 EV: Low

[0245] BE0 to 3_EV: Lo or High

[0246] Moreover, the pieces of data included in the second serial data are set as follows.

[0247] D0_OD: High

[0248] D1_OD: High

[0249] BE0 to 2_OD: Low

[0250] BE3_OD: High

[0251] Then, in the case where the _HE goes to Low after the rising of the LS, total of two drive circuits **103** corresponding to the heaters Seg20 and Seg 52 are activated, and the heaters Seg20 and Seg 52 are driven.

[0252] Moreover, for example, in the case where four heaters Seg0, Seg20, Seg32, and Seg 52 are to be selected, the values illustrated in FIG. **15** are set in the first serial data and the second serial data. Specifically, the pieces of data included in the first serial data are set as follows.

[0253] D0_EV: High

[0254] D1_EV: High

[0255] BE0 to 3_EV: Low

[0256] Moreover, the pieces of data included in the second serial data are set as follows.

[0257] D0_OD: High

[0258] D1_OD: High

[0259] BE0 to 2_OD: Low

[0260] BE3_OD: High

[0261] Then, in the case where the _HE goes to Low after the rising of the LS, total of four drive circuits **103** corresponding to the heaters Seg0, Seg20, Seg32, and Seg 52 are activated, and the heaters Seg0, Seg20, Seg32, and Seg 52 are driven.

[0262] Zero to four drive circuits **103** can be activated depending on the number of pieces of data set to High among the D0_EV, the D1_EV, the D0_OD, and the D1_OD, and zero to four heaters can be driven in response to this activation.

[0263] Since the heaters arranged in one array at high density can be driven as described above as in the above-mentioned embodiments, the liquid ejection head in which the ejection element substrate **100** is downsized can be provided, and the liquid ejection head including this ejection element substrate **100** can be also downsized. Moreover, arranging the heater array and the ink supply ports closer to one another can further downsize the ejection element substrate **100**, and can also achieve high-speed drive of the heaters.

Fifth Embodiment

[0264] An ejection element substrate **100** according to a fifth embodiment has the basic configuration as illustrated in FIG. **12**, like the ejection element substrate **100** according to the fourth embodiment.

[0265] FIG. **16** is a circuit diagram illustrating circuits mounted in the ejection element substrate **100** according to the fifth embodiment. FIG. **16** particularly illustrates details of the first selection circuit **105** and the second selection circuit **106**.

[0266] FIG. **16** is the circuit diagram illustrating the circuits mounted in the ejection element substrate **100** according to the fifth embodiment. A basic configuration of the circuits mounted in the ejection element substrate **100** according to the fifth embodiment is the same as the basic configuration of the circuits mounted in the ejection element substrate **100** according to the second embodiment. The connection relationship between the drive circuits **103** and the heaters **101** in the ejection element substrate **100** according to the second embodiment is as illustrated in FIG. **1**, and is illustrated also in FIG. **6**. Meanwhile, a connection relationship between the drive circuits **103** and the heaters **101** in the ejection element substrate **100** according to the fifth embodiment is as illustrated in FIG. **12**, and is illustrated also in FIG. **16**. As apparent from FIGS. **1**, **6**, **12**, and **16**, the ejection element substrate **100** according to the fifth embodiment is different from the ejection element substrate **100** according to the second embodiment only in the connection relationship between the drive circuits **103** and the heaters **101**. Accordingly, only the connection relationship between the drive circuits **103** and the heaters **101** is explained, and overlapping explanation of the common parts is omitted.

[0267] 16 drive circuits **103** selectable by the selector **215** are connected to the heaters Seg0 to 3, Seg8 to 11, Seg16 to 19, and Seg24 to 27, respectively. 16 drive circuits **103** selectable by the selector **216** are connected to the heaters Seg32 to 35, Seg40 to 43, Seg48 to 51, and Seg56 to 59, respectively. 16 drive circuits **103** selectable by the selector **235** are connected to the heaters Seg4 to 7, Seg12 to 15, Seg20 to 23, and Seg28 to 31, respectively. 16 drive circuits **103** selectable by the selector **236** are connected to the heaters Seg36 to 39, Seg44 to 47, Seg52 to 55, and Seg60 to 63, respectively.

[0268] An operation of the circuits mounted in the ejection element substrate **100** according to the fifth embodiment is the same as the operation of the circuits mounted in the ejection element substrate **100** according to the second embodiment. Since this operation is already explained with reference to FIG. **7**, overlapping explanation is omitted.

[0269] FIG. **17** illustrates a table illustrating a relationship between the selection control data inputted into the selection circuits **105** and **106** illustrated in FIG. **16** and the heaters **101** selected by these selection circuits **105** and **106**. In this case, the selection control data is included in the serial data. The BE0 EV, the BE1 EV, the BE2_EV, and the BE3 EV included in the serial data form the first selection control data encoded in binary. The D0 EV and the D1 EV included in the serial data are the pieces of enable data for enabling the selectors **215** and **216**, respectively. The BE0_OD, the BE1_OD, the BE2_OD, and the BE3_OD included in the serial data form the second selection control data encoded in binary. The D0_OD and the D1_OD included in the serial data are the pieces of enable data for enabling the selectors **235** and **236**, respectively.

[0270] For example, in the case where only one heater Seg20 is to be selected, the values illustrated in FIG. **17** are set in the serial data. Specifically, the pieces of data included in the serial data are set as follows.

[0271] D0 EV: Low

[0272] D1 EV: Low

[0273] BE0 to 3_EV: Low or High

[0274] Moreover, the other pieces of data included in the serial data are set as follows.

[0275] D0_OD: High

[0276] D1_OD: Low

[0277] BE0 to 2_OD: Low

[0278] BE3_OD: High

[0279] Then, in the case where the _HE goes to Low after the rising of the LS, one drive circuit **103** corresponding to the heater Seg20 is activated, and the heater Seg20 is driven.

[0280] Moreover, for example, in the case where only two heaters Seg20 and Seg 52 are to be selected, the values illustrated in FIG. **17** are set in the serial data. Specifically, the pieces of data included in the serial data are set as follows.

[0281] D0 EV: Low

[0282] D1 EV: Low

[0283] BE0 to 3_EV: Low or High

[0284] Moreover, the other pieces of data included in the serial data are set as follows.

[0285] D0_OD: High

[0286] D1_OD: High

[0287] BE0 to 2_OD: Low

[0288] BE3_OD: High

[0289] Then, in the case where the _HE goes to Low after the rising of the LS, two drive circuits **103** corresponding to the heaters Seg20 and Seg 52 are activated, and the heaters Seg20 and Seg 52 are driven.

[0290] Moreover, for example, in the case where four heaters Seg0, Seg20, Seg32, and Seg 52 are to be selected, the values illustrated in FIG. **17** are set in the serial data. Specifically, the pieces of data included in the serial data are set as follows.

[0291] D0_EV: High

[0292] D1_EV: High

[0293] BE0 to 3_EV: Low

[0294] Moreover, the other pieces of data included in the serial data are set as follows.

[0295] D0_OD: High

[0296] D1_OD: High

[0297] BE0 to 2_OD: Low

[0298] BE3_OD: High

[0299] Then, in the case where the _HE goes to Low after the rising of the LS, total of four drive circuits **103** corresponding to the heaters Seg0, Seg20, Seg32, and Seg 52 are activated, and the heaters Seg0, Seg20, Seg32, and Seg 52 are driven.

[0300] Zero to four drive circuits **103** can be activated depending on the number of pieces of data set to High among the D0_EV, the D1_EV, the D0_OD, and the D1_OD, and zero to four heaters can be driven in response to this activation.

[0301] Since the heaters arranged in one array at high density can be driven as described above as in the above-mentioned embodiments, the liquid ejection head in which the ejection element substrate **100** is downsized can be provided, and the liquid ejection head including this ejection element substrate **100** can be also downsized. Moreover, since the number of electrodes for data input can be reduced, it is possible to reduce the sealing regions for preventing invasion of the ink into the electrodes and improve reliability, as in the second and third embodiments. Furthermore, arranging the heater array and the ink supply ports closer to one another can further downsize the ejection element substrate **100**, and can also achieve high-speed drive of the heaters as in the fourth embodiment.

Sixth Embodiment

[0302] The ejection element substrate **100** according to a sixth embodiment has the basic configuration as illustrated in FIG. **12**, like the ejection element substrate **100** according to the fourth embodiment.

[0303] FIG. **18** is a circuit diagram illustrating circuits mounted in the ejection element substrate **100** according to the sixth embodiment. FIG. **18** particularly illustrates details of the first selection circuit **105** and the second selection circuit **106**.

[0304] FIG. **18** is the circuit diagram illustrating the circuits mounted in the ejection element substrate **100** according to the sixth embodiment. A basic configuration of the circuits mounted in the ejection element substrate **100** according to the sixth embodiment is the same as the basic configuration of the circuits mounted in the ejection element substrate **100** according to the third embodiment. The connection relationship between the drive circuits **103** and the heaters **101** in the ejection element substrate **100** according to the third embodiment is as illustrated in FIG. **1**, and is illustrated also in FIG. **9**. Meanwhile, a connection relationship between the drive circuits **103** and the heaters **101** in the ejection element substrate **100** according to the sixth embodiment is as illustrated in FIG. **12**, and is illustrated also in FIG. **18**. As apparent from FIGS. **1**, **9**, **12**, and **18**, the ejection element substrate **100** according to the sixth embodiment is different from the ejection element substrate **100** according to the third embodiment only in the connection relationship between the drive circuits **103** and the heaters **101**. Accordingly, only the connection relationship between the drive circuits **103** and the heaters **101** is explained, and overlapping explanation of the common parts is omitted.

[0305] Eight drive circuits **103** selectable by the selector **416** are connected to the heaters Seg0 to 3 and Seg8 to 11, respectively. Eight drive circuits **103** selectable by the selector **420** are connected to the heaters Seg16 to 19 and Seg24 to 27, respectively. Eight drive circuits **103** selectable by the selector **415** are connected to the heaters Seg32 to 35 and Seg40 to 43, respectively. Eight drive circuits **103** selectable by the selector **419** are connected to the heaters Seg48 to 51 and Seg56 to 59, respectively. Eight drive circuits **103** selectable by the selector **432** are connected to the heaters Seg4 to 7 and Seg12 to 15, respectively. Eight drive circuits **103** selectable by the selector **428** are connected to the heaters Seg20 to 23 and Seg28 to 31, respectively. Eight drive circuits **103** selectable by the selector **431** are connected to the heaters Seg36 to 39 and Seg44 to 47, respectively. Eight drive circuits **103** selectable by the selector **427** are connected to the heaters Seg52 to 55 and Seg60 to 63, respectively.

[0306] An operation of the circuits mounted in the ejection element substrate **100** according to the sixth embodiment is the same as the operation of the circuits mounted in the ejection element substrate **100** according to the third embodiment. Since this operation is already explained with reference to FIG. **10**, overlapping explanation is omitted.

[0307] A relationship between the selection control data inputted into the selection circuits **105** and **106** illustrated in FIG. **18** and the heaters **101** selected by these selection circuits **105** and **106** is the same as that in the third embodiment, and is explained with reference to FIG. **11**. Accordingly, overlapping explanation is omitted.

[0308] Driving the heaters arranged in one array at high density and reducing the DFFs, the LTs, and the decoders as described above can provide a liquid ejection head in which the ejection element substrate **100** is downsized.

[0309] Moreover, arranging the heater array and the ink supply ports closer to one another can further downsize the ejection element substrate **100**, and can also achieve high-speed drive of the heaters.

[0310] Since the heaters arranged in one array at high density can be driven as described above as in the above-mentioned embodiments, the liquid ejection head in which the ejection element substrate **100** is downsized can be provided, and the liquid ejection head including this ejection element substrate **100** can be also downsized.

[0311] Moreover, the DFFs, the LTs, and the decoders can be reduced from those in the first, second, fourth, and fifth embodiments. The ejection element substrate **100** can be further downsized, and the liquid ejection head including this ejection element substrate **100** can be also downsized. Furthermore, since the number of electrodes for data input can be reduced as in the second, third, and fifth embodiments, it is possible to reduce the sealing regions for preventing invasion of the ink into the electrodes and improve reliability.

[0312] Moreover, arranging the heater array and the ink supply ports closer to one another as in the fourth and fifth embodiments can further downsize the ejection element substrate **100**, and can also achieve high-speed drive of the heaters.

OTHER EMBODIMENTS

[0313] The ejection element substrates **100** according to the above-mentioned embodiments include two ink supply port arrays **102***a* and **102***b* for one heater array **101***a*. However, the present disclosure is not limited to this, and the ejection element substrate **100** may include only one ink supply port array **102***a* for one heater array **101***a* as illustrated in FIGS. **19** and **20**.

[0314] In the ejection element substrates **100** according to the above-mentioned embodiments, the multiple heaters **101** included in the heater array **101***a* are linearly aligned in one array having a main direction. However, the present disclosure is not limited to this, and as illustrated in FIG. **21**, the multiple heaters **101** included in the heater array **101***a* may be aligned in a staggered pattern in two arrays having the main direction and aligned in a sub-direction orthogonal to the main direction. In this case, an interval between the two arrays forming the staggered pattern is preferably smaller than twice the pitch of the heaters **101** in the direction (Y direction) of the heater array **101***a*, more preferably, equal to or smaller than the pitch.

[0315] In the ejection element substrate **100** according to the above-mentioned embodiments, the arrangement intervals (pitch) of the heaters are 1,200 dpi. However, the present disclosure is not limited to this, and the arrangement intervals of the heaters may have another value.

[0316] In the ejection element substrate **100** according to the above-mentioned embodiments, the first selection circuit **105** and the second selection circuit **106** include the selectors, the ANDs, the DFFs, the LTs, and the decoders. However, the present disclosure is not limited to this, and any or all of the ANDs, the DFFs, the LTs, and the decoders may be omitted from the first selection circuit **105** and the second selection circuit **106**. In this case, the omitted circuits may be provided inside or outside the ejection element substrate **100** as circuits separate from the first selection circuit **105** and the second selection circuit **106**.

[0317] In the ejection element substrate **100** according to the above-mentioned embodiments, the heaters to be selected by the first selection circuit **105** and the heaters to be selected by the second selection circuit **106** are arranged alternately, or the set of four heaters to be selected by the first selection circuit **105** and the set of four heaters to be selected by the second selection circuit **106** are arranged alternately. However, the present disclosure is not limited to this, and a set of any other integer number of heaters to be selected by the first selection circuit **105** and a set of the integer number of heaters to be selected by the second selection circuit **106** may be alternately arranged, the integer number being one or more. For example, a set of two or eight heaters to be selected by the first selection circuit **105** and a set of two or eight heaters to be selected by the second selection circuit **106** may be arranged alternately.

[0318] The ejection element substrate **100** according to the above-mentioned embodiments only have one system of the circuit as illustrated in FIG. **1**, **12**, or **19**. However, the present disclosure is not limited to this, and the ejection element substrate **100** may have multiple systems of the circuit as illustrated in FIG. **1**, **12**, or **19**.

[0319] The present disclosure includes the following configuration.

[Configuration 1]

[0320] An ejection element substrate comprising:

- **[0321]** a plurality of first energy generation elements;
- **[0322]** a plurality of first drive circuits configured to drive the plurality of first energy generation elements, respectively;
- **[0323]** a first selection circuit configured to select some of the first drive circuits from the plurality of first drive circuits and causes the selected first drive circuits to drive some of the first energy generation elements corresponding to the selected the first drive circuits, respectively;
- **[0324]** a plurality of second energy generation elements;
- **[0325]** a plurality of second drive circuits configured to drive the plurality of second energy generation elements, respectively; and
- **[0326]** a second selection circuit configured to select some of the second drive circuits from the plurality of second drive circuits and cause the selected second drive circuits to drive some of the second energy generation elements corresponding to the selected second drive circuits, respectively, wherein the plurality of first drive circuits are arranged side by side in a first direction, and the plurality of second drive circuits are arranged side by side in a first direction, and the plurality of first energy generation elements and the plurality of second energy generation elements are arranged side by side in a region between a region where the plurality of first drive circuits are arranged and a region where the plurality of second drive circuits are arranged such that the first direction is a main direction.

[Configuration 2]

[0327] The ejection element substrate according to configuration 1, wherein

- **[0328]** the second selection circuit includes as many second selectors as a maximum total number of the second energy generation elements simultaneously selectable by the second selection circuit, and each of the second selectors is controlled by third control data for selecting the second selectors to be enabled and fourth control data for designating the second energy generation elements to be selected by the enabled second selectors.

[Configuration 3]

[0329] The ejection element substrate according to configuration 2, further comprising a second conversion circuit configured to parallelize the third control data and the fourth control data included in series data inputted to the ejection element substrate.

[Configuration 4]

[0330] The ejection element substrate according to configuration 2 or 3, wherein

[0331] the fourth control data is encoded in binary, and

[0332] the ejection element substrate further comprises a second decoder configured to expand the fourth control data encoded in binary to data individually designating the second energy generation elements to be selected by the enabled second selectors.

[Configuration 5]

[0333] The ejection element substrate according to configuration 1, wherein

[0334] the first selection circuit includes as many third selectors as a maximum total number of the first energy generation elements simultaneously selectable by the first selection circuit,

[0335] the second selection circuit includes as many fourth selectors as a maximum total number of the second energy generation elements simultaneously selectable by the second selection circuit,

[0336] each of the third selectors is controlled by fifth control data for selecting the third selectors to be enabled and sixth control data for designating the first energy generation elements to be selected by the enabled third selectors,

[0337] each of the fourth selectors is controlled by seventh control data for selecting the fourth selectors to be enabled and eighth control data for designating the second energy generation elements to be selected by the enabled fourth selectors, and

[0338] the ejection element substrate further comprises:

[0339] a third conversion circuit configured to parallelize the fifth control data and the sixth control data included in series data inputted to the ejection element substrate; and

[0340] a fourth conversion circuit configured to parallelize the seventh control data and the eighth control data included in the series data.

[Configuration 6]

[0341] The ejection element substrate according to configuration 5, wherein

[0342] the sixth control data is encoded in binary, and

[0343] the ejection element substrate further comprises a third decoder configured to expand the sixth control data encoded in binary to data individually designating the first energy generation elements to be selected by the enabled third selectors.

[Configuration 7]

[0344] The ejection element substrate according to configuration 5 or 6, wherein

[0345] the eighth control data is encoded in binary, and

[0346] the ejection element substrate further comprises a fourth decoder configured to expand the eighth control data encoded in binary to data individually designating the second energy generation elements to be selected by the enabled fourth selectors.

[Configuration 8]

[0347] The ejection element substrate according to configuration 1, wherein

[0348] a set of the first selection circuit and the second selection circuit is capable of selecting at least one of the first energy generation elements, at least one of the second energy generation elements, or both of the at least one of the first energy generation elements and the at least one of the second energy generation elements,

[0349] the first selection circuit further includes as many fifth selectors as a maximum total number of the first energy generation elements and the second energy generation elements simultaneously selectable by the set of the first selection circuit and the second selection circuit,

[0350] the second selection circuit further includes as many sixth selectors as a maximum total number, and

[0351] each of the fifth selectors and the sixth selectors are controlled by ninth control data for selecting a set of the fifth selectors and the sixth selectors to be enabled and tenth control data for designating the first energy generation elements or the second energy generation elements to be selected by the set of the enabled fifth selectors and the enabled sixth selectors.

[Configuration 9]

[0352] The ejection element substrate according to configuration 8, further comprising a fifth conversion circuit configured to parallelize the ninth control data and the tenth control data included in series data inputted to the ejection element substrate.

[Configuration 10]

[0353] The ejection element substrate according to configuration 8 or 9, wherein

[0354] the tenth control data is encoded in binary, and the ejection element substrate further comprises a fifth decoder configured to expand the tenth control data encoded in binary to data individually designating the first energy generation elements or the second energy generation elements to be selected by the set of the enabled fifth selectors and the enabled sixth selectors.

[0355] According to the present disclosure, the area of the ejection element substrate can be reduced.

[0356] While the present invention has been described with reference to exemplary embodiments, it is to be understood that the invention is not limited to the disclosed exemplary embodiments. The scope of the following claims is to be accorded the broadest interpretation so as to encompass all such modifications and equivalent structures and functions.

[0357] This application claims the benefit of Japanese Patent Application No. 2024-032205, filed Mar. 4, 2024, which is hereby incorporated by reference wherein in its entirety.

What is claimed is:

1. An ejection element substrate comprising:

a plurality of first energy generation elements;

a plurality of first drive circuits configured to drive the plurality of first energy generation elements, respectively;

a first selection circuit configured to select some of the first drive circuits from the plurality of first drive circuits and cause the selected first drive circuits to drive some of the first energy generation elements corresponding to the selected first drive circuits, respectively;

a plurality of second energy generation elements;

a plurality of second drive circuits configured to drive the plurality of second energy generation elements, respectively; and

a second selection circuit configured to select some of the second drive circuits from the plurality of second drive circuits and cause the selected second drive circuits to drive some of the second energy generation elements corresponding to the selected second drive circuits, respectively, wherein

the plurality of first drive circuits are arranged side by side in a first direction and the plurality of second drive circuits are arranged side by side along the first direction, and

the plurality of first energy generation elements and the plurality of second energy generation elements are arranged side by side in a region between a region where the plurality of first drive circuits are arranged and a region where the plurality of second drive circuits are arranged such that a set of an integer number of the first energy generation elements and a set of the integer number of the second energy generation elements are arranged alternately with the first direction being a main direction, the integer number being one or more.

2. The ejection element substrate according to claim **1**, wherein the plurality of first energy generation elements and the plurality of second energy generation elements are arranged to be linearly aligned in one array having the first direction.

3. The ejection element substrate according to claim **1**, wherein the plurality of first energy generation elements and the plurality of second energy generation elements are arranged to be aligned in a staggered pattern of two arrays having the first direction.

4. The ejection element substrate according to claim **1**, wherein the first selection circuit is arranged in a region that is on a side on which the region where the plurality of first drive circuits are arranged is present, with respect to the region where the plurality of first energy generation elements and the plurality of second energy generation elements are arranged side by side, and that is farther away from the region where the plurality of first energy generation elements and the plurality of second energy generation elements are arranged side by side than the region where the plurality of first drive circuits are arranged.

5. The ejection element substrate according to claim **1**, wherein the second selection circuit is arranged in a region that is on a side on which the region where the plurality of second drive circuits are arranged is present, with respect to the region where the plurality of first energy generation elements and the plurality of second energy generation elements are arranged side by side, and that is farther away from the region where the plurality of first energy generation elements and the plurality of second energy generation elements are arranged side by side than the region where the plurality of second drive circuits are arranged.

6. The ejection element substrate according to claim **1**, wherein

a plurality of first ink supply ports are opened side by side in a region between the region where the plurality of first energy generation elements and the plurality of second energy generation elements are arranged side by side and the region where the plurality of first drive circuits are arranged, and

a plurality of second ink supply ports are opened side by side in a region between the region where the plurality of first energy generation elements and the plurality of second energy generation elements are arranged side by side and the region where the plurality of second drive circuits are arranged.

7. The ejection element substrate according to claim **6**, wherein

the plurality of first energy generation elements and the plurality of second energy generation elements are arranged side by side linearly in one array having the first direction, or arranged side by side in a staggered pattern of two arrays having the first direction, and

each of at least some of the first ink supply ports included in the plurality of first ink supply ports and at least some of the second ink supply ports included in the plurality of second ink supply ports are arranged at the same position in the first direction.

8. The ejection element substrate according to claim **6**, wherein

the plurality of first energy generation elements and the plurality of second energy generation elements are arranged side by side linearly in one array having the first direction, or arranged side by side in a staggered pattern of two arrays having the first direction, and

at least some of the first ink supply ports included in the plurality of first ink supply ports and at least some of the second ink supply ports included in the plurality of second ink supply ports are arranged alternately in the first direction.

9. The ejection element substrate according to claim **8**, wherein wiring lines that connect the first energy generation elements and the first drive circuits to one another and that are partially arranged in a region between each adjacent two of the first ink supply ports extend linearly in a second direction orthogonal to the first direction, in the region between each adjacent two of the first ink supply ports and a region between the first energy generation elements and the region between each adjacent two of the first ink supply ports.

10. The ejection element substrate according to claim **8**, wherein wiring lines that connect the second energy generation elements and the second drive circuits to one another and that are partially arranged in a region between each adjacent two of the second ink supply ports extend linearly in a second direction orthogonal to the first direction, in the region between each adjacent two of the second ink supply ports and a region between the second energy generation elements and the region between each adjacent two of the second ink supply ports.

11. The ejection element substrate according to claim **1**, wherein a plurality of third ink supply ports are opened side by side in a region between the region where the plurality of first energy generation elements and the plurality of second energy generation elements are arranged side by side and the region where the plurality of first drive circuits are arranged.

12. The ejection element substrate according to claim **1**, wherein

the plurality of first drive circuits are arranged side by side at a first pitch in the first direction, the plurality of second drive circuits are arranged side by side at the first pitch in the first direction,

the plurality of first energy generation elements and the plurality of second energy generation elements are arranged side by side at a second pitch in the first direction, and

the first pitch is twice the second pitch.

13. The ejection element substrate according to claim **1**, wherein

the some of the first energy generation elements selected by the first selection circuit is one or more of the first energy generation elements, and

the some of the second energy generation elements selected by the second selection circuit is one or more of the second energy generation elements.

14. The ejection element substrate according to claim **1**, wherein

the first selection circuit includes as many first selectors as a maximum total number of the first energy generation elements simultaneously selectable by the first selection circuit, and

each of the first selectors is controlled by first control data for selecting the first selectors to be enabled and second control data for designating the first energy generation elements to be selected by the enabled first selectors.

15. The ejection element substrate according to claim **14**, further comprising a first conversion circuit configured to parallelize the first control data and the second control data included in series data inputted to the ejection element substrate.

16. The ejection element substrate according to claim **14**, wherein

the second control data is encoded in binary, and

the ejection element substrate further comprises a first decoder configured to expand the second control data encoded in binary to data individually designating the first energy generation elements to be selected by the enabled first selectors.

17. The ejection element substrate according to claim **1**, wherein

the second selection circuit includes as many second selectors as a maximum total number of the second energy generation elements simultaneously selectable by the second selection circuit, and

each of the second selectors is controlled by third control data for selecting the second selectors to be enabled and fourth control data for designating the second energy generation elements to be selected by the enabled second selectors.

18. The ejection element substrate according to claim **1**, wherein

the first selection circuit includes as many third selectors as a maximum total number of the first energy generation elements simultaneously selectable by the first selection circuit,

the second selection circuit includes as many fourth selectors as a maximum total number of the second energy generation elements simultaneously selectable by the second selection circuit,

each of the third selectors is controlled by fifth control data for selecting the third selectors to be enabled and sixth control data for designating the first energy generation elements to be selected by the enabled third selectors,

each of the fourth selectors is controlled by seventh control data for selecting the fourth selectors to be enabled and eighth control data for designating the second energy generation elements to be selected by the enabled fourth selectors, and

the ejection element substrate further comprises:

a third conversion circuit configured to parallelize the fifth control data and the sixth control data included in series data inputted to the ejection element substrate; and

a fourth conversion circuit configured to parallelize the seventh control data and the eighth control data included in the series data.

19. The ejection element substrate according to claim **1**, wherein

a set of the first selection circuit and the second selection circuit is capable of selecting at least one of the first energy generation elements, at least one of the second energy generation elements, or both of the at least one of the first energy generation elements and the at least one of the second energy generation elements,

the first selection circuit further includes as many fifth selectors as a maximum total number of the first energy generation elements and the second energy generation elements simultaneously selectable by the set of the first selection circuit and the second selection circuit,

the second selection circuit further includes as many sixth selectors as the maximum total number, and

each of the fifth selectors and the sixth selectors is controlled by ninth control data for selecting a set of the fifth selectors and the sixth selectors to be enabled and tenth control data for designating the first energy generation elements or the second energy generation elements to be selected by the set of the enabled fifth selectors and the enabled sixth selectors.

20. A liquid ejection head comprising:

an ejection element substrate; and

an orifice plate attached to the ejection element substrate, wherein

the ejection element substrate includes:

a plurality of first energy generation elements;

a plurality of first drive circuits configured to drive the plurality of first energy generation elements, respectively;

a first selection circuit configured to select some of the first drive circuits from the plurality of first drive circuits and cause the selected first drive circuits to drive some of the first energy generation elements corresponding to the selected the first drive circuits, respectively;

a plurality of second energy generation elements;

a plurality of second drive circuits configured to drive the plurality of second energy generation elements, respectively; and

a second selection circuit configured to select some of the second drive circuits from the plurality of second drive circuits and cause the selected second drive circuits to drive some of the second energy generation elements corresponding to the selected second drive circuits, respectively,

the plurality of first drive circuits are arranged side by side along a first direction, and the plurality of second drive circuits are arranged side by side along the first direction, the plurality of first energy generation elements and the plurality of second energy generation elements are arranged side by side in a region between a region where the plurality of first drive circuits are arranged and a region where the plurality of second drive circuits are arranged such that the first direction is a main direction,

a boundary portion between the ejection element substrate and the orifice plate is provided with

a first common flow path communicating with a plurality of first ink supply ports,

a second common flow path communicating with a plurality of second ink supply ports,

a plurality of first individual flow paths communicating with the first common flow path,

a plurality of second individual flow paths communicating with the second common flow path, and

a plurality of pressure chambers communicating with the plurality of first individual flow paths, respectively, and with the plurality of second individual flow paths, respectively,

the orifice plate includes a plurality of ejection ports that allow the plurality of pressure chambers to communicate with an outside, and

the energy generation elements are arranged at positions where the energy generation elements face the ejection ports of the pressure chambers, respectively.

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